When mutualism turns costly: abiotic stress shifts endophyte benefits in cool-season grasses

Jacob K. Moutouama *[1], Josh Fowler[2], Akiem Gough[3], Julia Martin[3], Chris Oxley[3], Karl Schrader[3], Karis Williams[3], and Tom E.X. Miller [3]

[1]Department of Botany, University of British Columbia, Vancouver, BC, Canada
[2]Josh has a new CU affiliation
[3]Program in Ecology and Evolutionary Biology, Department of BioSciences, Rice University, Houston, TX USA

Running header: Stress and Grass–Endophyte Interactions

Keywords: demography, mutualism, parasitism, plant–fungi interactions, range limits, stress gradient hypothesis

Submitted to: *Ecology letters* (Letter)

Data accessibility statement: If the paper is accepted, all data and computer scripts supporting the results will be archived in a Zenodo package, with the DOI included at the end of the article. During peer review, our data and code[1] (Stan and R) is available at `https://github.com/jmoutouama/endo-range-limits`.

Conflict of interest statement: None.

Acknowledgments This research was supported by National Science Foundation Division of Environmental Biology award 2208857.

Authorship statement:

Abstract:
Main Text:
Figures: 5
Tables: 1
References: X

*Corresponding author: jmoutouama@gmail.com

[1]*If the data are not on github as csv's then I would create a temporary dropbox link so that reviewers could reproduce the analysis.*

Abstract

Benefits of plant-microbe symbioses are predicted to increase under environmental stress, but the relative roles of abiotic and biotic factors as modifiers of symbiotic interaction outcomes remain unclear. We studied how the independent and combined effects of aridity and herbivory modify the demographic effects of seed-transmitted fungal endophytes on three native grass host species using geographically distributed field experiments along a precipitation gradient, crossed with herbivore exclusion. Host fitness declined under increasing abiotic stress, and endophyte-symbiotic grasses often fared worse than non-symbiotic ones[2], suggesting that interactions that are beneficial under benign conditions can become costly under stress. Herbivory had no detectable effect on host demography or endophyte-mediated outcomes.[3] Overall, abiotic stress was the primary driver of mutualism outcomes between plants and fungal symbionts, but in the opposite direction as predicted by theory and prior work. These results highlight that symbiotic fungi can become liabilities under stress, with implications for the dynamics of plant-fungal mutualisms under global change.

[2] *I know space is tight but can we make this a more specific or quantitative summary of the results?*
[3] *I think this is not true in the new results...?*

Introduction

Plant-microbe[4] symbioses are widespread and ecologically important. These interactions are famously context-dependent, where the direction and strength of the interaction outcome depend on the environment in which it occurs (Bronstein, 1994; Hoeksema and Bruna, 2015). [5] For example, microbial symbionts that are beneficial to host-plants under stressful conditions[6] may become neutral or parasitic under benign conditions (Giauque et al., 2019). Variation in the nature of biotic and abiotic stress is a key driver of these context-dependent outcomes.

Because microbial symbioses are expected to become more beneficial in stressful conditions, they can substantially enhance host resilience to environmental stress. Under biotic stress, such as herbivory, microbial symbiosis can benefit host plants by facilitating the production of secondary compounds that deter feeding or exert direct toxicity, thereby reducing herbivore growth, survival, and oviposition (Atala et al., 2022; Bastias et al., 2017; Vega, 2008). Similarly, under abiotic stress, such as low precipitation, nutrient limitation, or high temperature, symbiotic microbes can enhance host tolerance through a variety of mechanisms (Afkhami and Rudgers, 2008; Clay and Schardl, 2002). Symbionts including mycorrhizal fungi, rhizobia, and endophytic bacteria can improve water and nutrient acquisition by increasing root surface area, facilitating nutrient mobilization, or promoting nitrogen fixation, allowing plants to maintain growth and metabolic function under drought or poor soil conditions (Amijee et al., 1989). In addition, many symbiotic microbes stimulate the production of antioxidant enzymes, which reduce oxidative damage caused by stressors like drought, salinity, and heat (Ruiz-lazona et al., 1996). By improving resource acquisition, modulating plant physiology, and enhancing stress-responsive metabolism, these symbiotic interactions help host plants buffer against environmental stochasticity and extreme conditions across a wide range of ecosystems (Fowler et al., 2023).

Although microbial symbioses often enhance plant performance, they can also impose substantial costs, particularly in resource-limited environments. Maintaining symbionts requires hosts to allocate valuable carbon and nutrients to sustain their microbial partners. For instance, mycorrhizal fungi may require up to 20% of photosynthetically fixed carbon[7] (Soudzilovskaia et al., 2015; Thirkell et al., 2020), and under abiotic stress, this investment

[4] *For Eco Letters I would keep this more generally, so start with symbiosis generally and not plant-microbe specifically. Aphid symbioses often involve protection from heat stress and defense against enemies, so it's an animal parallel to what we see in plants. For a broad journal, you will want to emphasize the breadth of these ideas.*

[5] *To keep this very general, you may want to bring in the SGH here, and emphasize the point that stressful conditions are expected to strengthen positive interactions (the opposite of what you found).*

[6] *Since you are citing a specific example here, I would be more specific about what the stressful conditions were.*

[7] *This is cool, would be great to do something like this with grass endophytes.*

can outweigh the benefits of the symbiosis (Bronstein, 2001; Johnson et al., 1997), reducing host survival, growth, reproduction, or competitive ability (Klironomos, 2003). Similarly, vertically transmitted endophytes can shift from mutualistic to antagonistic interactions when they produce stromata that abort host inflorescences or impose energetic costs (Saikkonen et al., 2004). Experimental studies further show that under low nutrient or water conditions, endophyte-infected grasses may produce fewer tillers and lower total biomass compared to endophyte-free plants, reflecting the costs of sharing limited resources (Ahlholm et al., 2002).[8]

Despite growing interest in the role of symbiosis in shaping host demography, few studies have tested how abiotic and biotic stressors synergistically influence the demographic effects of symbionts along environmental gradients. Most existing work has been conducted in controlled or common garden experiments, where, for example, herbivory but not drought reduced plant growth, and no interaction between the two stressors was observed (Emery et al., 2010).[9] Studying the combined effects of abiotic and biotic stressors under natural conditions is particularly important in the context of global change (Louthan et al., 2015), as shifting climate patterns and altered herbivore pressures may change the balance of costs and benefits in plant–microbe interactions, potentially influencing species distributions and ecosystem function. Understanding how abiotic stress, biotic stress, and microbial symbiosis interact to influence plant performance requires experiments that jointly vary all three factors. However, to our knowledge, no such comprehensive study has been conducted in native grass species[10]. Here, we address this gap by examining the effects of biotic and abiotic stress on grass-endophyte symbioses in three grass species spanning a natural range-core/range-edge precipitation gradient.

Fungal endophytes of plants provide a tractable system for studying the ecological effects and context-dependence of host-symbiont interactions, as host status (symbiotic vs symbiont-free) can be experimentally manipulated. A well-studied example is the association between cool-season grasses and seed-transmitted fungal endophytes in the genus *Epichloë* (Clay et al., 2005; Oberhofer et al., 2014; Saikkonen et al., 2010; Schardl et al., 2004). Vertical transmission links the fitness of host-plants with that of fungal partners, and is expected to favor the evolution of host-symbiont mutualism (cite). Endophyte-symbiotic grasses often perform better than endophyte-free grasses under water limitation, although benefits can

[8] *This is the opposite of the expectations you set up above, that symbionts can become more beneficial under drought stress, which might be a little confusing or disorienting for readers. So I think you need to acknowledge that tension and work it into the "story" of the Introduction. One way to do this would be setting this up as a competing hypothesis, e.g. "While stress is hypothesized to strengthen the benefits of symbiotic microbes, growing evidence suggests that it could also elevate costs of symbiosis."*

[9] *I would add the symbiont component of this example, which is the main point you are trying to make. Also, if this was a single-species study, I would highlight that, maybe by identifying the species.*

[10] *Is this last bit important? Is the sentence still true if we cut it?*

vary with host, endophyte species, or genotype (Saikkonen et al., 1998).[11] Beyond drought tolerance, infection frequency itself is influenced by herbivory and environmental conditions, including long-term variation in precipitation and temperature (Clay et al., 2005; Fowler et al., 2025; Rudgers et al., 2016).[12] Even when endophytes do not enhance survival under stress, they may still boost host fitness by promoting growth and reproduction (Yule et al., 2013). More generally, prior work has shown that effects of endophyte symbiosis may vary in magnitude and even sign across host vital rates (Rudgers et al. 2012)[13], which argues for considering multiple life history pathways. Together, these features make grass–*Epichloë* symbioses an ideal system for investigating how abiotic stressors, such as drought, interact with biotic stressors, like herbivory, to determine the outcome of symbiosis. [14]

Across an aridity gradient in the south-central United States, we investigated how endophyte symbiosis affects multiple host vital rates under varying abiotic and biotic stress. This gradient, from mesic conditions in the east to increasingly arid conditions in the west, provides a biologically realistic stress gradient, and coincides with the westernmost range limits of many eastern North American grass species. At each of seven sites along the gradient, we manipulated vertebrate and invertebrate herbivory (allowing herbivore access versus excluding herbivores) to test its interaction with abiotic stress. We also manipulated endophyte presence, creating endophyte-free and endophyte-symbiotic individuals[15], to examine effects of *Epichloë* symbiosis in three species of native cool-season grasses.

We addressed two questions: (i) under combined abiotic and biotic stress, do endophytes provide fitness benefits or impose costs on their host plants? (ii) which factors (abiotic vs. biotic) determine whether the grass–endophyte interaction benefits or harms the host?[16] We predicted that endophytes would enhance host performance under low precipitation by mitigating drought stress (Fig. 1A; stress-amelioration scenario). Specifically, in fenced plots with reduced herbivory, endophyte benefits are expected to be smaller because the contribution of symbionts to host performance is less critical when biotic stress is low. In contrast, under

[11] *Decunta et al. 2021 is a newer reference.*

[12] *This sentence mentions several modifying factors but feels a little unfocused. I think you can sharpen the point you are trying to make here.*

[13] *https://doi.org/10.1890/11-0689.1*

[14] *Somewhere in this paragraph, which I think is meant to give a quick intro to the grass-endo system, I would say a bit more about the costs of hosting endophytes (plant carbon and risk of horizontal transmission). I would also say more in this paragraph about fungal alkaloids as the mechanism of herbivore deterrence, and cite some of the work on both vertebrate and invertebrate herbivores.*

[15] *I would soften this, because based on the analysis we did it is only in AGHY that we can be confident that our E- are experimentally disinfected. So overall it was a mix of disinfected E- and naturally occurring E-. It will be important to provide those results in the supplement.*

[16] *I don't really understand the difference between these two questions. Maybe we can sharpen this? Also, it is worth considering whether you want to include data on fungal stroma, which is a direct measure of one type of cost. We only ever saw this in AGHY.*

high abiotic stress combined with herbivore presence, the grass–endophyte symbiosis may become less mutualistic, as herbivory can suppress or interfere with the positive effects of endophytes and could even impose a net cost (Fig. 1B; inhibition scenario).[17] Furthermore, while both abiotic and biotic stress are expected to influence the outcome of endophyte-grass symbiosis, we anticipate that abiotic stress will be the primary driver of mutualistic benefits, with biotic stress modulating the magnitude or direction of those benefits.[18]

Materials and methods

Study system

[19] Our study focused on three cool-season perennial grasses native to woodland and prairie habitats of eastern North America: *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis* (Poaceae, subfamily Pooideae) (Shaw, 2011). *A. hyemalis* is a small, short-lived perennial that germinates from autumn to late winter and typically flowers between March and July. *E. virginicus* is a larger, long-lived species that begins flowering as early as March or April and continues throughout the summer. *P. autumnalis* has a similar growth form to *E. virginicus* and flowers from late spring through summer. *Agrostis hyemalis* hosts the endophyte *Epichloë amarillans* (White Jr, 1994), *E. virginicus* typically hosts *Epichloë elymi* (Liu, 2023), and *P. autumnalis* hosts *Epichloë PauTG-1* or *E. typhina subsp. Poae* (Gundel et al., 2020). Like other *Epichloë* species, these endophytes are primarily transmitted vertically through seeds and are additionally capable of horizontal transmission via sexual stromata on inflorescences or asexual conidia on leaf surfaces (Clay and Schardl, 2002; Gundel et al., 2020; Leuchtmann et al., 2014). However, horizontal transmission is very rare and was not observed in our study system.[20]

Source populations and symbiont removal

Plants used in our experiment were derived from natural (source) populations of each species (Fig. 2, Table S1). Prevalence of fungal endophytes in these natural populations ranged from XX% to XX% endophyte-positive hosts (NEW SUPP TABLE). Seeds from these source populations (from ca. 30 plants per population) were either heat-treated or left untreated,

[17] *I don't really understand where this prediction comes from. Fungal alkaloids have known anti-herbivore properties, so all else equal I would expect stronger mutualistic effects under herbivore access, which is what you imply in the preceding sentence. I don't see the mechanism for how herbivores might "interfere" with positive endo effects.*

[18] *The rationale for this expectation is not totally clear to me.*

[19] *I think this section should also say something about the herbivores in this system.*

[20] *This is not true. There were a good number of stroma observations on AGHY, and it would be possible to include those data here if you want.*

intended to produce symbiont-free (E^{-}) and symbiotic (E^{+}) plants from the same genetic lineages. For *Elymus virginicus* and *Poa autumnalis*, heat-treated seeds were placed in a drying over at 60°C for approximately five days. For *Agrostis hyemalis*, seeds were heat-treated in a water bath at 60°C for 12 minutes. [21] All seeds, both heat-treated and non-heat-treated, were surface-sterilized with 10% bleach solution then germinated on agar plates. Seedlings were transferred to potting soil and grown in the Rice University greenhouse. Once these plants were sufficiently large in size they were vegetatively propagated, with clonal identities tracked across individuals. [22] Before transfer to the field experiment, we confirmed endophyte status (E^{+} or E^{-}) of all seedlings using either microscopy or an immunoblot assay. [23] For microscopy, peels of the inner leaf sheaths were stained with aniline blue lactic acid and viewed under a compound microscope at 200x–400x to identify fungal hyphae [24]. The immunoblot assay (Phytoscreen field tiller endophyte detection kit, Agrinostics Ltd. Co.) uses monoclonal antibodies that target proteins of *Epichloë* spp. and a chromagen to visually indicate presence or absence (Koh et al., 2006). Both methods yield similar detection rates.[25] For genotypes that were vegetatively propagated, we tested the endophyte status of a single clonal replicate and assume that its status applies equally to all clones of a given source genotype. Greenhouse-raised plants were planted into the field experiment in February-March 2023.

Common garden experimental design and climate data

To examine the effects of endophyte symbiosis across an abiotic stress gradient, we established common garden experiments with E+ and E- hosts at seven sites along an aridity gradient from Louisiana to central Texas, US (Fig. 2, Table S2).[26] From west to east, mean annual precipitation increases more than three-fold across these sites, while mean annual temperature is relatively consistent (Fig. S1).[27] For this reason, we focus on precipitation as the primary abiotic factor

[21] *We should talk about this. You need more detail here – I have tried adding some but there is more to do. They were all treated but not all treatments were successful, which we need to report. And there were a few different methods used on different species. This should all go in the supplement. It is also not true that heat treatment has no effect on seed viability, which is why we tried to use seeds that were multiple generations removed from the heat treatment.*

[22] *Add a sentence giving basic descriptive statistics of how many genotypes and clones were used for each species.*

[23] *Realized numbers of E+ and E- per species per population should be provided in a suppl table.*

[24] *cite*

[25] *What is the evidence for this statement?*

[26] *Figure comments: can you spell out 'precipitation' rather than 'P'? Change gray dots to 'GBIF occurence'. Change 'Garden' to 'Experimental site'. In D, I would order sites from west to east, rather than precip rank. Legend needs to describe panels E-F. Unclear what the green and red colors in E are meant to show - this may get confused for E+/E-. I like the herbivory figure – it would be good to see this broken up by species, maybe as a supp figure.*

[27] *Unclear what you are showing here. Are these 30-year normals or realized weather? You say annual mean precip but this looks exactly like Figure 2D.*

that varied across sites. This geographic gradient overlaps and exceeds the natural range edges of our focal species, ensuring realistic exposure to climatic and biotic conditions (Fig. 2A,B, C).

To characterize climate at these sites during the study period (2023-2025), we collected daily temperature and precipitation data from PRISM (Hart and Bell, 2015). These daily values were used to calculate the mean temperature (°C) and cumulative precipitation (mm) for each year, spanning from the time plants were transplanted to when demographic data were collected.[28] During our study period, realized precipitation at these sites corresponded well to 30-year normals, with wetter eastern sites and drier western sites (Fig. 2D).

At each site along the gradient, we established 24 square plots (1.5 m $\times$ 1.5 m), eight per host species. Within each site, each plot was randomly assigned to either herbivore access or herbivore exclusion (Fig. 2E). The herbivory treatment targeted both vertebrate and invertebrate hebivores: exclusion plots had fencing on four sides and were sprayed with Sevin insecticide (active ingredient Zeta-Cypermethrin) 2-3 times per growing season following the manufacturer's protocol, and access plots were fenced on two sides to control for any fence effect. To ensure that our herbivory treatment reflected the level of herbivory experienced by individual plants, we counted the number of damaged plants per plot and estimated the proportion of damaged plants per site and herbivore treatment. Our results confirmed that the herbivory treatment was effective (Fig. 2F).[29]

Each plot was planted with 15 individuals, a mix of E+ and E- host-plants. Initial endophyte prevalence varied from 20% (3 E+, 12 E-) to 80% (12 E+, 3 E-); this aspect of the experimental design was intended to address a different question about change in symbiont prevalence across generations, and is not further considered here. For each species at each site along the precipitation gradient, we planted a total of 60 E+ and 60 E- host-plants, half of those experiencing herbivore access and half experiencing herbivore exclusion.

[30]

Demographic data collection

We collected demographic data, including survival, growth, and reproduction, during the flowering season of each species: *Agrostis hyemalis* and *Poa autumnalis* were censused in May and *Elymus virginicus* was censused in June of 2023, 2024, and 2025. For each individual,

[28] *I am a little confused here because the figure says this is cumulative precipitation over the study period, but here you say you calculated precip per year, so is this an average over years? And when you say 'year', is this the calendar year or the census year?*

[29] *I recommend moving the methods to data collection methods below (you will also need to define what you mean by "damage" and move the result to results).*

[30] *Kerrville and Sonora sites were destroyed and replanted in 2024. That should be described here. Also, POAU was not planted in Sonora. The map is accurate but you need to describe that here.*

survival was recorded as alive or dead and, for surviving plants, size was measured as the count of tillers. Growth was quantified as the log ratio of tiller count between consecutive censuses (i.e., $\log(\text{tiller}_{t+1}/\text{tiller}_t)$) for plants that were alive in years t and $t+1$. We recorded the number of inflorescences per plant for all three species and counted the number of spikelets on up to three inflorescences from reproducing individuals of *Poa autumnalis* and *Elymus virginicus*. In *A. hyemalis*, one spikelet yields one seed, while in *P. autumnalis* and *E. virginicus* a single spikelet can produce 2-6 seeds.

Assessing whether endophytes confer benefits or impose costs under abiotic and biotic stress

To assess whether endophytes confer benefits or impose costs under abiotic and biotic stress, we first developed two candidate models for each vital rate: one quadratic and one monotonic. All models shared the same hierarchical structure and were applied to survival, growth, and fertility. Survival was modeled with a Bernoulli distribution, growth with a Gaussian distribution, the number of inflorescences with a zero-inflated negative binomial distribution, and the number of spikelets with a negative binomial distribution. We modeled all vital rates (survival, growth, and fertility) using species-specific fixed effects for symbiosis (endophyte presence), precipitation, and herbivore access, including all relevant two and three-way interactions. Each observation i corresponds to a plant in a given year; individuals can appear in multiple years. For simplicity, we omit the full subscript notation for species, plot, population, and site-year, which is explicitly included in the Stan model. We present the quadratic model here (Eq. 1) because it encompasses the linear model as a special case.

$$
\begin{aligned}
\mu_i = {} & \beta_0[Sp_i] + \beta_P[Sp_i]P_i + \beta_{P^2}[Sp_i]P_i^2 + \beta_{endo}[Sp_i]endo_i + \beta_{herb}[Sp_i]herb_i \\
& + \beta_{endo\times P}[Sp_i]endo_i P_i + \beta_{herb\times P}[Sp_i]herb_i P_i \\
& + \beta_{herb\times endo}[Sp_i]herb_i endo_i + \beta_{endo\times herb\times P}[Sp_i]endo_i herb_i P_i \\
& + \phi_{\text{site}[i],Sp_i} + \omega_{\text{source}[i],Sp_i} + \rho_{\text{plot}[i],Sp_i}.
\end{aligned}
\tag{1}
$$

Here, P_i is total precipitation (mm) cumulated over the 12 months preceding the census, $endo_i$ indicates endophyte presence, and $herb_i$ indicates herbivore access. Random effects account for background variation at multiple hierarchical levels: $\phi_{\text{site}[i],Sp_i}$ captures site-to-site heterogeneity, $\omega_{\text{source}[i],Sp_i}$ captures variation associated with the provenance of transplants, and $\rho_{\text{plot}[i],Sp_i}$ captures plot-to-plot variation. Each species has its own random effect for each site, source, and plot, while the variance parameters (σ_{site}, σ_{source}, σ_{plot}) are shared across species. This hierarchical structure allows the model to "borrow strength" across species, improving estimates of fixed effects while still capturing species-specific responses to environmental gradient. We confirmed that models effectively captured key aspects of the grass demography data, including means, standard deviations, and quantiles using a posterior predictive check (Fig. S2).

Secondly, we compared quadratic and monotonic candidate models using leave-one-out cross-validation (LOOCV) to identify the best-supported model for each vital rate (Vehtari et al., 2017). LOOCV combines validation and training by iteratively holding out one observation as a test set while fitting the model on the remaining $n-1$ observations. This process is repeated n times, once for each observation, resulting in n fitted models (Silva and Zanella, 2024). The overall test error estimate is then obtained by averaging the prediction errors across all n models (Eq. 2) for each vital rate.

$$CV_n = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{2}$$

Although we evaluated quadratic models, LOOCV showed that they provided minimal improvement over monotonic models (Table S3). Therefore, all subsequent analyses used the monotonic model, which captures the observed patterns while maintaining model simplicity.

Finally, for each vital rate, precipitation was summarized at key quantiles (10th, 25th, 50th, 75th, and 90th percentiles) to represent the range of site-level abiotic stress. For each species (*Agrostis hyemalis*, *Elymus virginicus*, *Poa autumnalis*) and herbivory treatment (Fenced or Unfenced), we computed posterior predictions of survival probability for individuals with endophytes (E^+) and without endophytes (E^-) using the best Bayesian model outputs (Eq. 1). The difference ($\Delta = E^+ - E^-$) was calculated for each posterior sample to estimate median effects and 90% credible intervals. We also computed the posterior probability that $\Delta > 0$, representing the likelihood that endophytes confer benefits on vital rates under the given climate and herbivory conditions.

Drivers of endophyte effects on cool-season grass demography

To identify the main drivers of endophyte-mediated effects, we fitted three hierarchical Bayesian candidate models for each vital rate (survival, growth, inflorescence, and spikelet production; Supplementary Methods S.1.1). Each model included endophyte status as a baseline predictor and differed in whether it incorporated interactions with precipitation (Endophyte × Climate), herbivory (Endophyte × Herbivory), or both factors, including the three-way interaction (Endophyte × Climate × Herbivory). Random effects for site-year ($\phi_{\mathrm{site}[i],\mathrm{Sp}_i}$), source population ($\omega_{\mathrm{source}[i],\mathrm{Sp}_i}$), and plot ($\rho_{\mathrm{plot}[i],\mathrm{Sp}_i}$) captured hierarchical variation while accommodating species-specific responses. We compared candidate models using leave-one-out cross-validation (LOO-CV) (Vehtari et al., 2017), which estimates predictive accuracy by iteratively holding out each observation. Differences in expected log predictive density (ΔELPD), along with associated standard errors, were used to rank model performance.

All models were implemented in R (R Core Team, 2023) using Stan via the `rstan` interface (Stan Development Team, 2024).

Results

Endophyte effects on host fitness components across abiotic and biotic stress gradients

Across fitness components, *Epichloë* endophytes generally enhanced host performance under wetter conditions, whereas their effects were costly or occasionally neutral under drier conditions, consistent with the inhibition scenario (Fig. 1). Endophyte-mediated benefits were consistent regardless of herbivore access.

Grass-*Epichloë* symbiosis had mostly negative or neutral effects on survival at low precipitation, but increasingly enhanced survival under wetter conditions (Fig. 3; Fig. S3). In *Agrostis hyemalis* and *Elymus virginicus*, median $\Delta(E^{+}-E^{-})$ ranged from approximately -0.026 to 0.019 under low precipitation in unfenced plots ($\Pr(\Delta > 0) = 0.37$–0.61) and increased to 0.022–0.106 under high precipitation ($\Pr(\Delta > 0) = 0.70$–0.85; Table S4). In fenced plots, positive effects appeared only at the highest precipitation for *Agrostis* (median $\Delta = 0.011$–0.038, $\Pr(\Delta > 0) = 0.59$–0.72), while for *Elymus*, strong positive effects were evident at high precipitation (median $\Delta = 0.089$–0.113, $\Pr(\Delta > 0) = 0.91$–0.95). *Poa autumnalis* showed little survival benefit in unfenced plots across the gradient (median $\Delta = -0.056$ to -0.002, $\Pr(\Delta > 0) = 0.14$–0.43), but weakly positive effects emerged when herbivores were excluded (median $\Delta = 0.002$–0.026, $\Pr(\Delta > 0) = 0.57$–0.71).

Epichloë endophyte presence generally enhanced relative growth of individuals across the precipitation gradient, with benefits increasing under wetter conditions and not modulated by herbivory (Table S5). In *A. hyemalis*, effects were strongly positive even at moderate precipitation in unfenced plots (median $\Delta = 0.370$–0.385; $\Pr(\Delta > 0) = 0.946$–0.992), and similar trends were observed in fenced plots though slightly weaker (median $\Delta = 0.080$–0.262; $\Pr(\Delta > 0) = 0.661$–0.878). *E. virginicus* showed small positive or neutral effects under dry conditions (median $\Delta = 0.232$–0.259; $\Pr(\Delta > 0) = 0.869$–0.953), which became more pronounced at higher precipitation (median $\Delta = 0.124$–0.134; $\Pr(\Delta > 0) = 0.779$–0.790). In *P. autumnalis*, growth was context-dependent: unfenced plots shifted from negative to positive effects across the precipitation gradient (median $\Delta = -0.275$ to 0.144; $\Pr(\Delta > 0) = 0.169$–0.775), while fencing generally amplified benefits at low to moderate precipitation (median $\Delta = 0.020$–0.390; $\Pr(\Delta > 0) = 0.559$–0.927), but became negative at higher precipitation (median $\Delta = -0.262$ to -0.322; $\Pr(\Delta > 0) = 0.056$–0.072).

Epichloë endophytes consistently increased inflorescence production across the precipitation gradient (Fig. 5). By contrast, spikelet production was negatively affected by *Epichloë* under low precipitation but increased under wetter conditions, with effects varying among species and depending on herbivore exclusion (Fig. S6; Fig. S7). Inflorescence benefits in *A. hyemalis* and *E. virginicus* were minimal under dry conditions but increased with precipitation, with only minor influence from herbivore exclusion. In *P. autumnalis*, strong inflorescence benefits occurred under fencing at moderate-to-high precipitation (median $\Delta = 0.47$–1.54; $\Pr(\Delta > 0) = 0.96$–1.00; Table S6). Spikelet production showed a similar pattern: *E. virginicus* and *P. autumnalis* experienced weak or negative effects under low precipitation (median $\Delta = 0.10$–0.40; $\Pr(\Delta > 0) = 0.23$–0.40), shifting to positive effects at higher precipitation, with fencing amplifying these benefits across the gradient (median $\Delta = 0.95$–1.56; $\Pr(\Delta > 0) = 0.79$–0.99; Table S7).

Abiotic factors as the primary drivers of fitness components across the stress gradient

Grass-endophyte symbiosis, whether mutualistic or parasitic, was primarily modulated by climatic variation rather than by herbivore pressure. Across all vital rates (survival, growth, inflorescence, and spikelet production), models comparing different drivers of the endophyte–host interaction consistently identified the *Endophyte* $\times$ *Climate* model as the best predictor of host demography (Table 1). For instance, for survival, the *Endophyte* $\times$ *Climate* model outperformed the *Endophyte* $\times$ *Herbivory* model (ΔELPD $= -2.1$, SE = 2.8), and including the three-way *Endophyte* $\times$ *Climate* $\times$ *Herbivory* interaction further decreased predictive performance (ΔELPD $= -5.0$, SE = 1.9). Similar patterns were observed for growth, inflorescence, and spikelet production, indicating that climatic context is the primary driver of variation in the demographic consequences of endophyte symbiosis, whereas herbivory has weaker or context-dependent effects.

Discussion

Drawing on three years of field experiments that crossed endophyte infection (infected vs. free) with herbivory exclusion, and using Bayesian statistical models, we evaluated how biotic and abiotic stressors jointly shape the demographic effects of vertically transmitted Epichloë endophytes in three cool-season perennial grasses across a natural climatic gradient in the United States. Our results indicate that grass–endophyte symbioses generally enhance host performance under favorable (e.g. wetter) conditions, whereas their effects can be neutral or even detrimental under high abiotic stress, consistent with the inhibition scenario. Across species and fitness components, climatic variation emerged as the dominant driver of endophyte-mediated effects, while herbivory had a secondary or context-dependent influence. These results underscore that the outcome of grass–*Epichloë* symbiosis shifts along abiotic stress gradients (Bronstein, 1994; Chamberlain et al., 2014; Saikkonen et al., 1998), providing empirical support for environmentally contingent mutualism in a broadly distributed host–symbiont system.

Endophyte presence often enhances host performance under favorable or moderately stressful conditions, a pattern widely documented across grass–*Epichloë* systems (Clay, 1990). Endophytes can increase host growth, reproduction, and resistance to herbivores (Rodriguez et al., 2009), but these benefits are most apparent when the host has sufficient resources, such as water and nutrients, to support the fungus (Clay, 1988; Gibert et al., 2012). Mechanistically, the host must allocate carbon and nutrients to maintain the endophyte, so under resource-rich conditions, the costs of symbiosis are outweighed by the benefits, resulting in a net positive effect. In our field experiment, we observed higher growth and survival of endophyte-infected grasses in wetter sites, consistent with previously documented positive effects of endophytes under favorable conditions (Afkhami et al., 2014; Rudgers et al., 2020).

While endophyte infection enhanced host performance under favorable or moderately stressful conditions, its effects diminished and ultimately became costly under high abiotic stress, even in fenced plots where herbivores were excluded. This pattern indicates that drought-driven resource limitation can impose costs on the symbiosis independent of biotic stress (Bronstein, 2001; Cheplick et al., 1989; Johnson et al., 1997). By contrast, other studies show that the context-dependent benefits of endophytes are often mediated by herbivore pressure (Afkhami and Rudgers, 2009; Bastias et al., 2017). For instance, in *Bromus setifolia*, endophyte infection is favored under high herbivore pressure, when defensive benefits outweigh the costs, but is selected against when herbivory is low or other costs are elevated (Rodriguez et al., 2009). Our results suggest that similar context dependence can arise from abiotic factors, such as drought, independent of herbivory. Under low-precipitation conditions, reduced water availability constrains nutrient uptake and increases the relative metabolic burden of maintaining the endophyte, potentially tipping the cost–benefit balance against endophyte infected plants (Ahlholm et al., 2002; Bronstein, 2001; Faeth, 2002). In addition, severe environmental stress can activate plant defense pathways-such as elevated phytohormone production and reactive oxygen species (ROS) that may inhibit endophyte growth or reduce its functional contributions to the host (Bastías and Gundel, 2023; Eaton et al., 2011). Consequently, even in the absence of herbivores, endophyte-mediated benefits may become negative in resource-limited environments, consistent with the reduced performance of infected plants at the driest sites in our study. Although we found no evidence of infection-related costs for inflorescences across the precipitation gradient, spikelet production showed a small but consistent cost at the driest sites. This indicates that drought-driven resource limitation disproportionately affects certain reproductive components in infected plants, with different traits varying in their sensitivity to shifts in host–endophyte interactions (Cheplick and Faeth, 2009).

Across species and fitness components, climatic variation was the primary driver of endophyte-mediated effects, whereas herbivory played a secondary or context-dependent role. Precipitation strongly shaped host demography in our system, with drier sites constraining survival, growth and fertility. Contrary to the predictions of the stress gradient hypothesis (SGH), which suggests that biotic interactions should become more positive under stressful conditions, endophyte-mediated benefits were reduced under high abiotic stress (Lortie and Callaway, 2006; Louthan et al., 2015). Although most empirical tests of the SGH have focused on plant–plant interactions and have been developed and tested mainly by community ecologists (Bertness and Callaway, 1994), recent syntheses show that support for the SGH is actually strongest among microbial interactions (Hammarlund and Harcombe, 2019; Piccardi et al., 2019), with bacterial systems showing greater conformity to SGH predictions than any other kingdom (Adams et al., 2022). This contrast highlights that plant–microbe mutualisms, including fungal endophytes, may not follow the same stress-dependency patterns documented in plant community ecology, and that endophyte benefits can decline rather than intensify under severe environmental stress. A possible explanation for this opposite pattern is that the benefits of fungal endophytes are closely tied to host resource availability. Under high abiotic stress, host plants have limited resources for survival, growth, and reproduction, which can reduce the capacity of endophytes to confer benefits. Unlike plant–plant interactions, where neighbors can alleviate stress through mechanisms like shading or nutrient modification, endophytes depend on the host plant's internal physiological condition, such as its nutrient and water status, energy reserves, and overall health. When host resources are constrained, the costs of maintaining the symbiosis may outweigh its benefits, leading to reduced endophyte-mediated effects under high abiotic stress.

References

Adams, A. E., Besozzi, E. M., Shahrokhi, G., and Patten, M. A. (2022). A case for associational resistance: Apparent support for the stress gradient hypothesis varies with study system. *Ecology Letters*, 25(1):202–217.

Afkhami, M. E., McIntyre, P. J., and Strauss, S. Y. (2014). Mutualist-mediated effects on species' range limits across large geographic scales. *Ecology letters*, 17(10):1265–1273.

Afkhami, M. E. and Rudgers, J. A. (2008). Symbiosis lost: imperfect vertical transmission of fungal endophytes in grasses. *The American Naturalist*, 172(3):405–416.

Afkhami, M. E. and Rudgers, J. A. (2009). Endophyte-mediated resistance to herbivores depends on herbivore identity in the wild grass festuca subverticillata. *Environmental Entomology*, 38(4):1086–1095.

Ahlholm, J. U., Helander, M., Lehtimäki, S., Wäli, P., and Saikkonen, K. (2002). Vertically transmitted fungal endophytes: different responses of host-parasite systems to environmental conditions. *Oikos*, 99(1):173–183.

Amijee, F., Tinker, P., and Stribley, D. (1989). The development of endomycorrhizal root systems: Vii. a detailed study of effects of soil phosphorus on colonization. *New Phytologist*, 111(3):435–446.

Atala, C., Acuña-Rodríguez, I. S., Torres-Díaz, C., and Molina-Montenegro, M. A. (2022). Fungal endophytes improve the performance of host plants but do not eliminate the growth/defence trade-off. *New Phytologist*, 235(2).

Bastías, D. A. and Gundel, P. E. (2023). Plant stress responses compromise mutualisms with epichloë endophytes. *Journal of Experimental Botany*, 74(1):19–23.

Bastias, D. A., Martínez-Ghersa, M. A., Ballaré, C. L., and Gundel, P. E. (2017). Epichloë fungal endophytes and plant defenses: not just alkaloids. *Trends in Plant Science*, 22(11):939–948.

Bertness, M. D. and Callaway, R. (1994). Positive interactions in communities. *Trends in ecology & evolution*, 9(5):191–193.

Bronstein, J. L. (1994). Conditional outcomes in mutualistic interactions. *Trends in ecology & evolution*, 9(6):214–217.

Bronstein, J. L. (2001). The exploitation of mutualisms. *Ecology letters*, 4(3):277–287.

Chamberlain, S. A., Bronstein, J. L., and Rudgers, J. A. (2014). How context dependent are species interactions? *Ecology letters*, 17(7):881–890.

Cheplick, G., Clay, K., and Marks, S. (1989). Interactions between infection by endophytic fungi and nutrient limitation in the grasses lolium perenne and festuca arundinacea. *New Phytologist*, 111(1):89–97.

Cheplick, G. P. and Faeth, S. H. (2009). *Ecology and evolution of the grass-endophyte symbiosis*. OUP USA.

Clay, K. (1988). Fungal endophytes of grasses: a defensive mutualism between plants and fungi. *Ecology*, 69(1):10–16.

Clay, K. (1990). Fungal endophytes of grasses. *Annual review of Ecology and Systematics*, pages 275–297.

Clay, K., Holah, J., and Rudgers, J. A. (2005). Herbivores cause a rapid increase in hereditary symbiosis and alter plant community composition. *Proceedings of the National Academy of Sciences*, 102(35):12465–12470.

Clay, K. and Schardl, C. (2002). Evolutionary origins and ecological consequences of endophyte symbiosis with grasses. *the american naturalist*, 160(S4):S99–S127.

Eaton, C. J., Cox, M. P., and Scott, B. (2011). What triggers grass endophytes to switch from mutualism to pathogenism? *Plant Science*, 180(2):190–195.

Emery, S. M., Thompson, D., and Rudgers, J. A. (2010). Variation in endophyte symbiosis, herbivory and drought tolerance of ammophila breviligulata populations in the great lakes region. *The American Midland Naturalist*, 163(1):186–196.

Faeth, S. H. (2002). Are endophytic fungi defensive plant mutualists? *Oikos*, 98(1):25–36.

Fowler, J. C., Donald, M. L., Bronstein, J. L., and Miller, T. E. (2023). The geographic footprint of mutualism: How mutualists influence species' range limits. *Ecological Monographs*, 93(1):e1558.

Fowler, J. C., Moutouama, J., and Miller, T. E. (2025). Increasing prevalence of plant-fungal symbiosis across two centuries of environmental change. *Global Change Biology*, 31(11):e70577.

GBIF (2025a). Occurrence download for *Agrostis hyemalis*. Accessed 5 November 2025.

GBIF (2025b). Occurrence download for *Elymus virginicus*. Accessed 5 November 2025.

GBIF (2025c). Occurrence download for *Poa autumnalis*. Accessed 5 November 2025.

Giauque, H., Connor, E. W., and Hawkes, C. V. (2019). Endophyte traits relevant to stress tolerance, resource use and habitat of origin predict effects on host plants. *New Phytologist*, 221(4):2239–2249.

Gibert, A., Volaire, F., Barre, P., and Hazard, L. (2012). A fungal endophyte reinforces population adaptive differentiation in its host grass species. *New Phytologist*, 194(2):561–571.

Gundel, P. E., Sun, P., Charlton, N. D., Young, C. A., Miller, T. E., and Rudgers, J. A. (2020). Simulated folivory increases vertical transmission of fungal endophytes that deter herbivores and alter tolerance to herbivory in poa autumnalis. *Annals of Botany*, 125(6):981–991.

Hammarlund, S. P. and Harcombe, W. R. (2019). Refining the stress gradient hypothesis in a microbial community. *Proceedings of the National Academy of Sciences*, 116(32):15760–15762.

Hart, E. M. and Bell, K. (2015). *prism: Download data from the Oregon prism project*. R package version 0.0.6.

Hoeksema, J. D. and Bruna, E. M. (2015). Context-dependent outcomes of mutualistic interactions. *Mutualism*, 10:181–202.

Johnson, N. C., Graham, J. H., and Smith, F. A. (1997). Functioning of mycorrhizal associations along the mutualism–parasitism continuum. *The New Phytologist*, 135(4):575–585.

Klironomos, J. N. (2003). Variation in plant response to native and exotic arbuscular mycorrhizal fungi. *Ecology*, 84(9):2292–2301.

Koh, S., Vicari, M., Ball, J., Rakocevic, T., Zaheer, S., Hik, D., and Bazely, D. (2006). Rapid detection of fungal endophytes in grasses for large-scale studies. *Functional Ecology*, pages 736–742.

Leuchtmann, A., Bacon, C. W., Schardl, C. L., White Jr, J. F., and Tadych, M. (2014). Nomenclatural realignment of neotyphodium species with genus epichloë. *Mycologia*, 106(2):202–215.

Liu, M. (2023). Fungi canadenses no. 351: Epichloe elymi. *Canadian Journal of Plant Pathology*, 45(3):300–303.

Lortie, C. J. and Callaway, R. M. (2006). Re-analysis of meta-analysis: support for the stress-gradient hypothesis. *Journal of Ecology*, 94(1):7–16.

Louthan, A. M., Doak, D. F., and Angert, A. L. (2015). Where and when do species interactions set range limits? *Trends in ecology & evolution*, 30(12):780–792.

Oberhofer, M., Güsewell, S., and Leuchtmann, A. (2014). Effects of natural hybrid and non-hybrid epichloë endophytes on the response of hordelymus europaeus to drought stress. *New Phytologist*, 201(1):242–253.

Piccardi, P., Vessman, B., and Mitri, S. (2019). Toxicity drives facilitation between 4 bacterial species. *Proceedings of the National Academy of Sciences*, 116(32):15979–15984.

Porter, S. S., Bantay, R., Friel, C. A., Garoutte, A., Gdanetz, K., Ibarreta, K., Moore, B. M., Shetty, P., Siler, E., and Friesen, M. L. (2020). Beneficial microbes ameliorate abiotic and biotic sources of stress on plants. *Functional Ecology*, 34(10):2075–2086.

R Core Team (2023). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria.

Rodriguez, R., White Jr, J., Arnold, A. E., and Redman, a. R. a. (2009). Fungal endophytes: diversity and functional roles. *New phytologist*, 182(2):314–330.

Rudgers, J. A., Afkhami, M. E., Bell-Dereske, L., Chung, Y. A., Crawford, K. M., Kivlin, S. N., Mann, M. A., and Nuñez, M. A. (2020). Climate disruption of plant-microbe interactions. *Annual review of ecology, evolution, and systematics*, 51:561–586.

Rudgers, J. A., Fletcher, R. A., Olivas, E., Young, C. A., Charlton, N. D., Pearson, D. E., and Maron, J. L. (2016). Long-term ungulate exclusion reduces fungal symbiont prevalence in native grasslands. *Oecologia*, 181(4):1151–1161.

Ruiz-lazona, J. M., Azcón, R., and Palma, J. (1996). Superoxide dismutase activity in arbuscular mycorrhizal lactuca sativa plants subjected to drought stress. *New Phytologist*, 134(2):327–333.

Saikkonen, K., Faeth, S. H., Helander, M., and Sullivan, T. (1998). Fungal endophytes: a continuum of interactions with host plants. *Annual review of Ecology and Systematics*, 29(1):319–343.

Saikkonen, K., Saari, S., and Helander, M. (2010). Defensive mutualism between plants and endophytic fungi? *Fungal Diversity*, 41(1):101–113.

Saikkonen, K., Wäli, P., Helander, M., and Faeth, S. H. (2004). Evolution of endophyte–plant symbioses. *Trends in plant science*, 9(6):275–280.

Schardl, C. L., Leuchtmann, A., and Spiering, M. J. (2004). Symbioses of grasses with seedborne fungal endophytes. *Annu. Rev. Plant Biol.*, 55(1):315–340.

Shaw, R. B. (2011). *Guide to Texas grasses*. Texas A&M University Press.

Silva, L. A. and Zanella, G. (2024). Robust leave-one-out cross-validation for high-dimensional bayesian models. *Journal of the American Statistical Association*, 119(547):2369–2381.

Soudzilovskaia, N. A., van der Heijden, M. G., Cornelissen, J. H., Makarov, M. I., Onipchenko, V. G., Maslov, M. N., Akhmetzhanova, A. A., and van Bodegom, P. M. (2015). Quantitative assessment of the differential impacts of arbuscular and ectomycorrhiza on soil carbon cycling. *New Phytologist*, 208(1):280–293.

Stan Development Team (2024). RStan: the R interface to Stan. R package version 2.32.6.

Thirkell, T. J., Pastok, D., and Field, K. J. (2020). Carbon for nutrient exchange between arbuscular mycorrhizal fungi and wheat varies according to cultivar and changes in atmospheric carbon dioxide concentration. *Global change biology*, 26(3):1725–1738.

Vega, F. E. (2008). Insect pathology and fungal endophytes. *Journal of invertebrate pathology*, 98(3):277–279.

Vehtari, A., Gelman, A., and Gabry, J. (2017). Practical bayesian model evaluation using leave-one-out cross-validation and waic. *Statistics and computing*, 27:1413–1432.

White Jr, J. F. (1994). Endophyte-host associations in grasses. xx. structural and reproductive studies of epichloë amarillans sp. nov. and comparisons to e. typhina. *Mycologia*, 86(4):571–580.

Yule, K. M., Miller, T. E., and Rudgers, J. A. (2013). Costs, benefits, and loss of vertically transmitted symbionts affect host population dynamics. *Oikos*, 122(10):1512–1520.

Figures

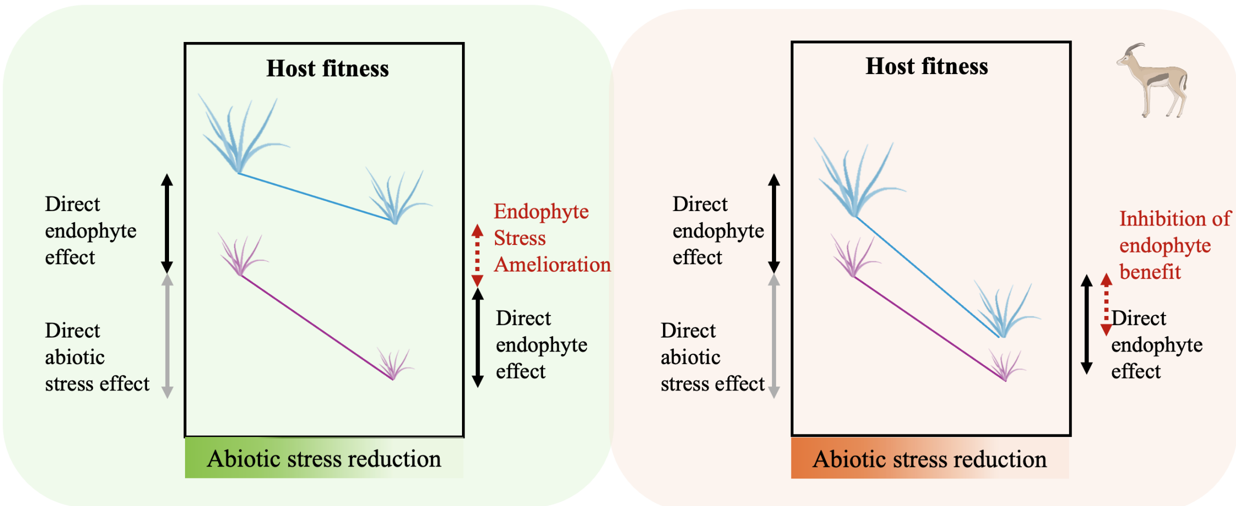

Figure 1: Conceptual framework illustrating the predicted effects of abiotic stress, biotic stress, and endophyte-mediated mutualism on host plant fitness (adapted from (Porter et al., 2020)). (a) **Stress-amelioration scenario:** Endophytes are predicted to enhance host performance under increasing abiotic stress (e.g., low precipitation) by mitigating drought-related physiological costs. Abiotic stress is the primary driver of host fitness, and endophyte-mediated benefits become more pronounced as stress intensifies. (b) **Inhibition scenario:** When abiotic stress is combined with an additional biotic stressor herbivory the positive direct effect of endophytes is expected to weaken. Under high combined stress (abiotic + biotic), herbivory suppresses or interferes with drought-mitigation benefits, reducing or eliminating the symbiotic advantage and potentially resulting in a net cost of hosting an endophyte. Colors are used solely for visualization: blue lines represent endophyte-infected plants and violet lines represent endophyte-free plants.

[31]

[31] *I have the PowerPoint for this, which you can find here.*

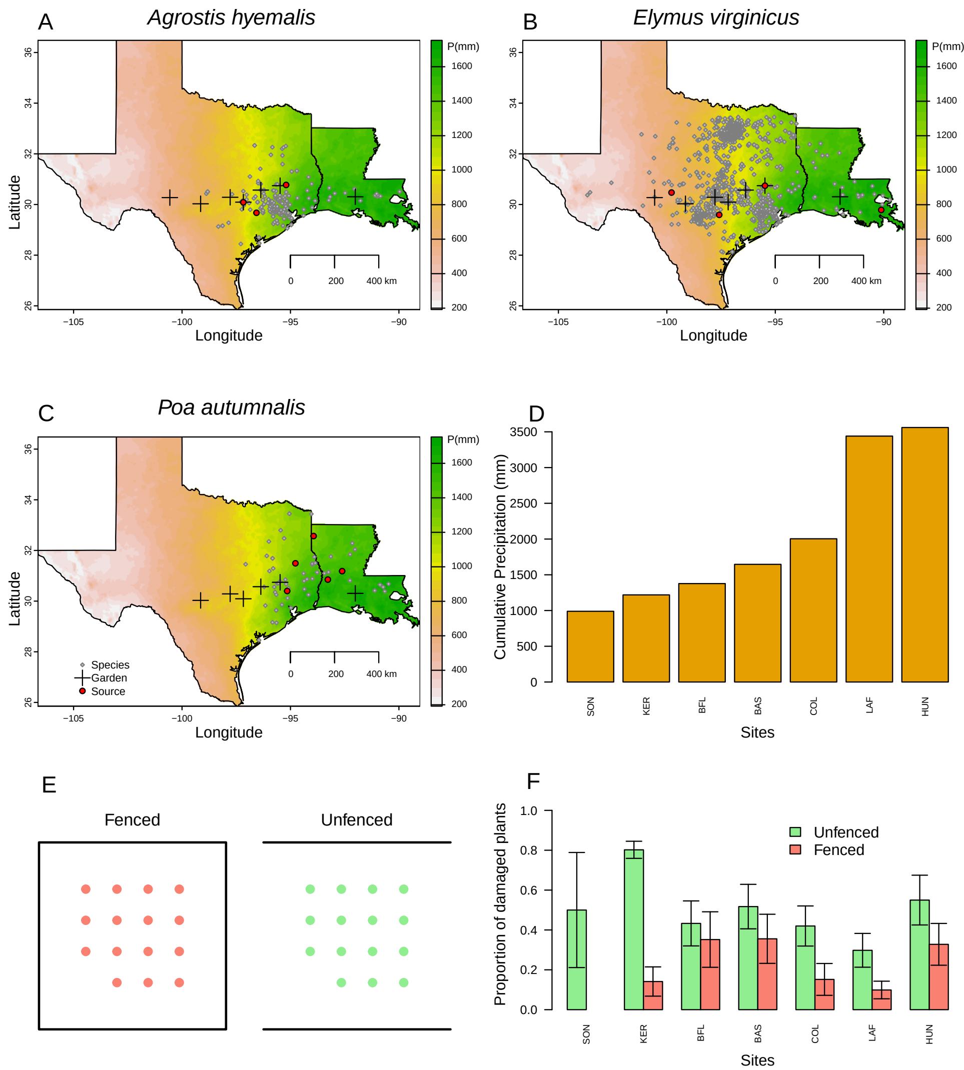

Figure 2: Distribution of common garden sites along a natural aridity gradient from Texas to Louisiana, US for (A) *Agrostis hyemalis*, (B) *Elymus virginicus*, and (C) *Poa autumnalis*. Climate data in A–C are based on 30-year normals of climate predictions (1993–2023) from PRISM data (Hart and Bell, 2015), which illustrate the precipitation gradient across sites (scale on the right is in mm. Red dots indicate the locations of source populations, while grey dots represent Global Biodiversity Information Facility (GBIF) occurrence records for each species within the study area (GBIF, 2025a,b,c). D) Cumulative precipitation during the study period (May 2023 – June 2025) at each site, ordered west to east. Values are also derived from PRISM climate data (Hart and Bell, 2015).

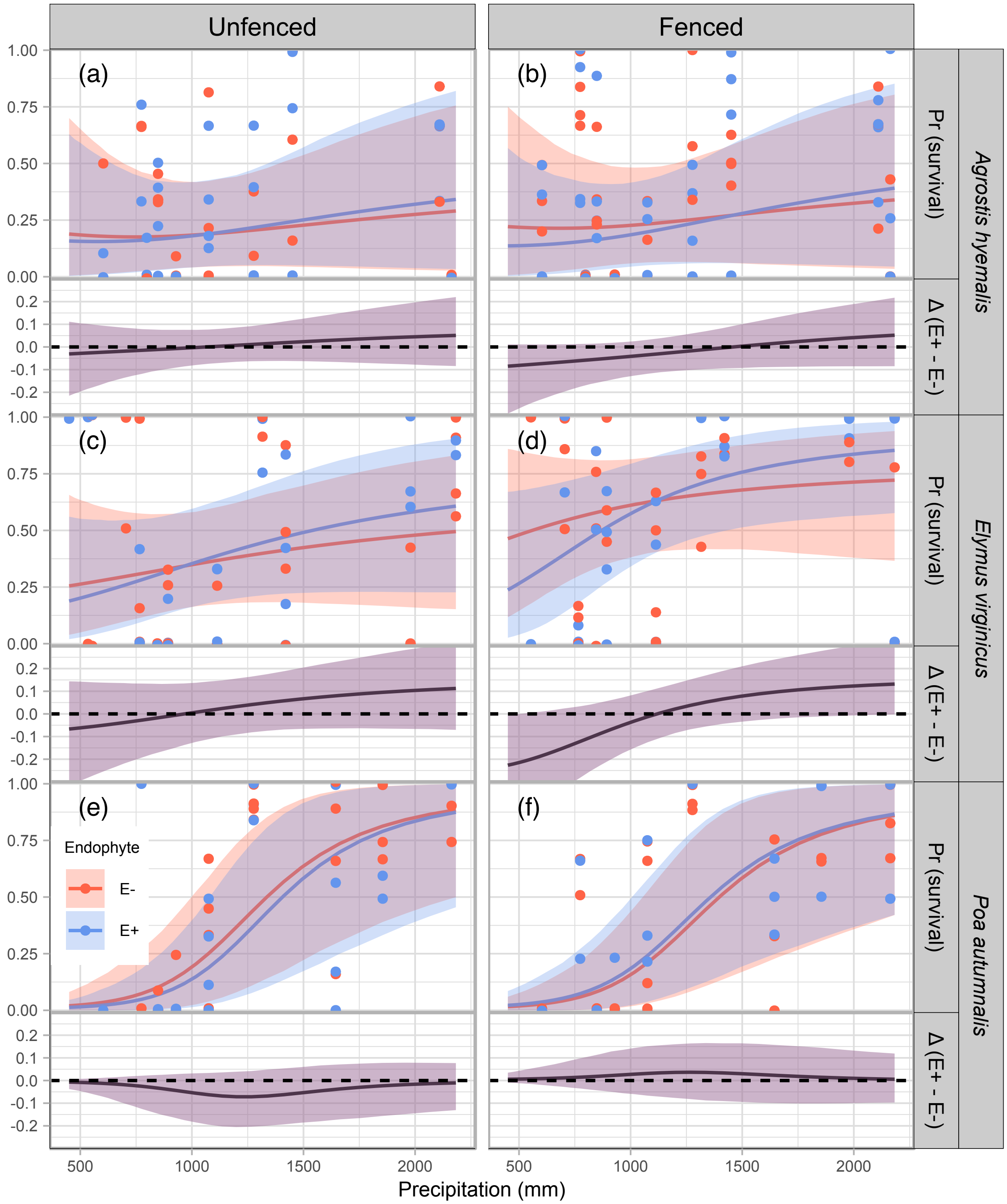

Figure 3: Predicted survival rate (top panels) and endophyte effect ($\Delta = E^{+} - E^{-}$, bottom panels) across precipitation gradients. Lines show posterior means for E^{+} (blue) and E^{-} (red), with shading indicating 90% credible intervals. Points represent mean observed survival per plot, jittered for visualization. The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

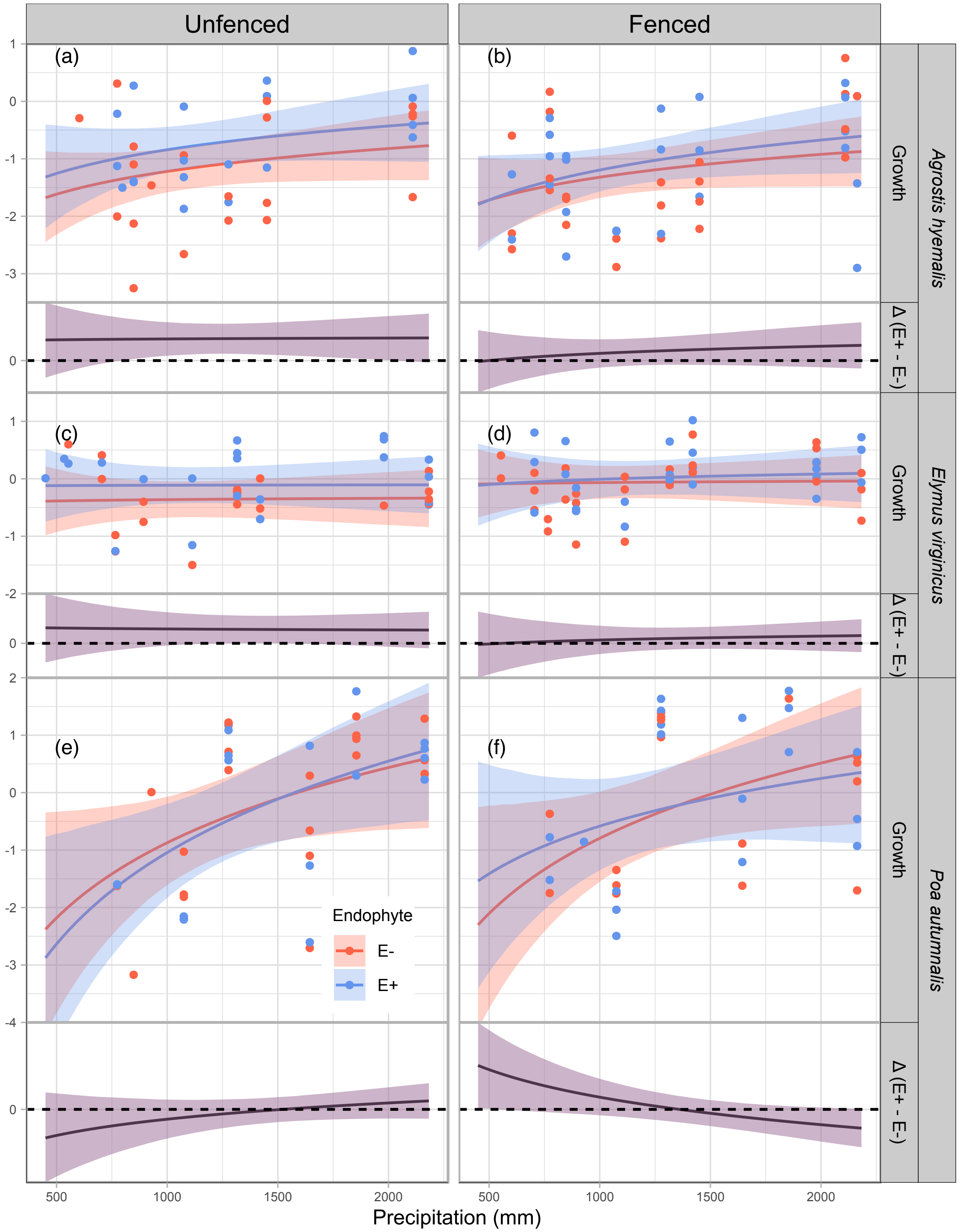

Figure 4: Predicted relative growth from year to the next year (top panels) and endophyte effect ($\Delta = E^+ - E^-$, bottom panels) across precipitation gradients. Lines show posterior means for E^+ (blue) and E^- (red), with shading indicating 90% credible intervals. Points represent mean observed growth per plot, jittered for visualization. The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

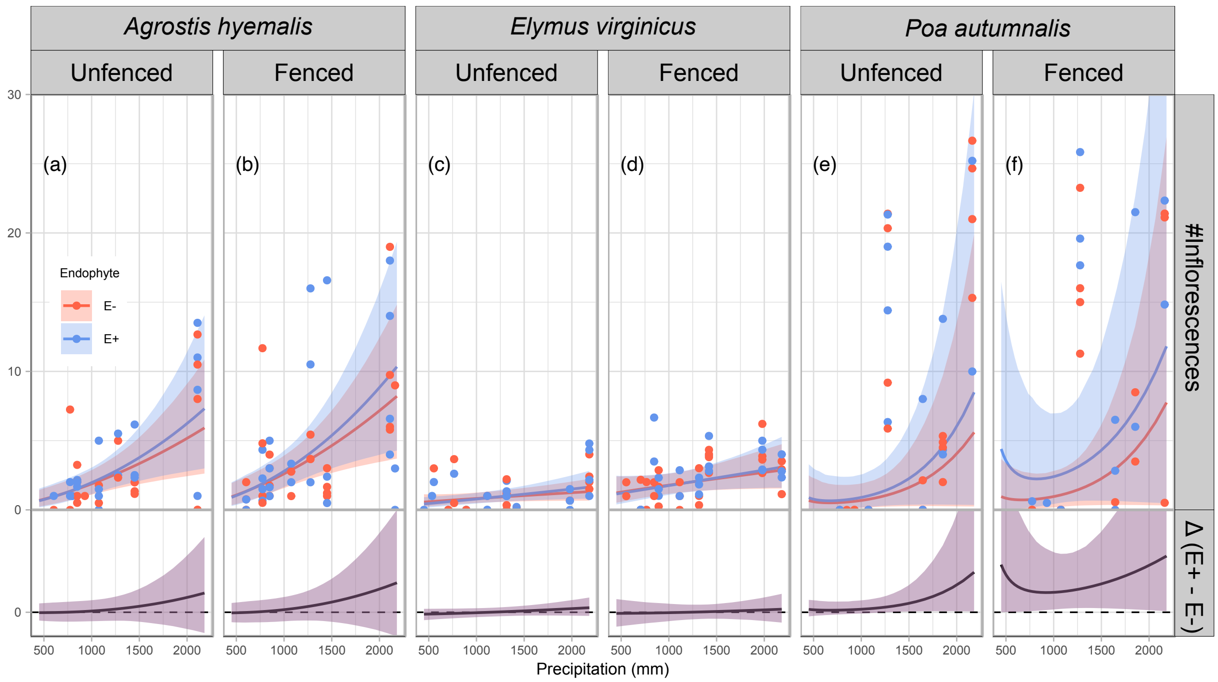

Figure 5: Predicted number of inflorescences (upper panels) and the difference between endophyte-infected (E^+) and endophyte-free (E^-) plants ($\Delta(E^+–E^-)$, lower panels) are shown across a gradient of precipitation. Shaded areas represent 90% credible intervals. Points indicate observed means, jittered for visualization. Panels are grouped by species and herbivory treatment (Fenced, Unfenced).

Tables

Table 1: Model comparison of drivers of grass- endophyte symbiosis across four fitness components (vital rates). ΔELPD indicates the difference in expected log predictive density relative to the best-fitting model; SE is the standard error of the difference.

Vital rate	**Model type**	**ΔELPD**	**SE**
Survival	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−2.11	2.81
	Endophyte × Climate × Herbivory	−4.98	1.89
Growth	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−0.19	2.81
	Endophyte × Climate × Herbivory	−1.81	2.75
Inflorescence	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−2.62	3.53
	Endophyte × Climate × Herbivory	−3.78	2.63
Spikelet	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−2.01	3.82
	Endophyte × Climate × Herbivory	−3.16	2.43

S.1 Supporting Information

S.1.1 Supporting Methods

Candidate models for abiotic and biotic drivers of cool-season grass species

1. **Endophyte × Climate model:** includes endophyte status (endo), precipitation (P), and their interaction:

$$\mu_i = \beta_0[\mathrm{Sp}_i] + \beta_{\mathrm{endo}}[\mathrm{Sp}_i] endo_i + \beta_P[\mathrm{Sp}_i] P_i + \beta_{\mathrm{endo}\times P}[\mathrm{Sp}_i] endo_i P_i$$
$$+ \phi_{\mathrm{site}[i],\mathrm{Sp}_i} + \omega_{\mathrm{source}[i],\mathrm{Sp}_i} + \rho_{\mathrm{plot}[i],\mathrm{Sp}_i}$$

2. **Endophyte × Herbivory model:** includes endophyte status (endo), herbivory (herb), and their interaction:

$$\mu_i = \beta_0[\mathrm{Sp}_i] + \beta_{\mathrm{endo}}[\mathrm{Sp}_i] endo_i + \beta_{\mathrm{herb}}[\mathrm{Sp}_i] herb_i + \beta_{\mathrm{endo}\times\mathrm{herb}}[\mathrm{Sp}_i] endo_i herb_i$$
$$+ \phi_{\mathrm{site}[i],\mathrm{Sp}_i} + \omega_{\mathrm{source}[i],\mathrm{Sp}_i} + \rho_{\mathrm{plot}[i],\mathrm{Sp}_i}$$

3. **Endophyte × Climate × Herbivory:** combines all predictors and interactions up to the three-way interaction:

$$\mu_i = \beta_0[\mathrm{Sp}_i] + \beta_P[\mathrm{Sp}_i] P_i + \beta_{\mathrm{endo}}[\mathrm{Sp}_i] endo_i + \beta_{\mathrm{herb}}[\mathrm{Sp}_i] herb_i$$
$$+ \beta_{\mathrm{endo}\times P}[\mathrm{Sp}_i] endo_i P_i + \beta_{\mathrm{herb}\times P}[\mathrm{Sp}_i] herb_i P_i + \beta_{\mathrm{herb}\times\mathrm{endo}}[\mathrm{Sp}_i] herb_i endo_i$$
$$+ \beta_{\mathrm{endo}\times\mathrm{herb}\times P}[\mathrm{Sp}_i] endo_i herb_i P_i + \phi_{\mathrm{site}[i],\mathrm{Sp}_i} + \omega_{\mathrm{source}[i],\mathrm{Sp}_i} + \rho_{\mathrm{plot}[i],\mathrm{Sp}_i}$$

In these models, the β coefficients represent species-specific effects. The intercept $\beta_0[\mathrm{Sp}_i]$ corresponds to the baseline value of the vital rate for species Sp_i in the absence of predictors. $\beta_{\mathrm{endo}}[\mathrm{Sp}_i]$ and $\beta_{\mathrm{herb}}[\mathrm{Sp}_i]$ describe the effects of endophyte presence and herbivory, respectively. The linear effect of precipitation is captured by $\beta_P[\mathrm{Sp}_i]$, while interactions are represented by $\beta_{\mathrm{endo}\times P}[\mathrm{Sp}_i]$, $\beta_{\mathrm{herb}\times P}[\mathrm{Sp}_i]$, $\beta_{\mathrm{endo}\times\mathrm{herb}}[\mathrm{Sp}_i]$, and the three-way interaction $\beta_{\mathrm{endo}\times\mathrm{herb}\times P}[\mathrm{Sp}_i]$. All models include hierarchical random effects for site-year, source population, and plot, with species-specific random effects $\phi_{\mathrm{site}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0, \sigma_{\mathrm{site}})$, $\omega_{\mathrm{source}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0, \sigma_{\mathrm{source}})$, and $\rho_{\mathrm{plot}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0, \sigma_{\mathrm{plot}})$.

S.1.2 Supporting Tables

Table S1: Geographic coordinates and brief descriptions of source populations for *Agrostis hyemalis* (AGHY), *Elymus virginicus* (ELVI), and *Poa autumnalis* (POAU).

Species	Population	Latitude	Longitude	Description
AGHY	SJC	30.771969	-95.200620	San Jacinto County, TX
AGHY	COHR	29.672833	-96.567517	Columbus, TX
AGHY	LPEF	30.085383	-97.172508	Lost Pines
POAU	LA7	30.849660	-93.276772	DeRidder, LA
POAU	LA1	32.568125	-93.932976	Jean Lafitte National Historical Park and Preserve[32]
POAU	LA2	31.183040	-92.616969	Jean Lafitte National Historical Park and Preserve
POAU	SFAEF	31.492158	-94.772200	Stephen F. Austin Experimental Forest, TX
POAU	WB	30.399346	-95.150026	Sam Houston National Forest
ELVI	HUNT	30.741797	-95.474508	Sam Houston State University Field Station
ELVI	SHS[33]	30.741797	-95.474508	Sam Houston State University Field Station
ELVI	RL5	30.471717	-99.783803	Texas Tech University Field Station, Junction
ELVI	PALM	29.591623	-97.584781	Palmetto State Park, TX
ELVI	JLP	29.785105	-90.114933	Jean Lafitte Park, outside of New Orleans

Table S2: Common garden sites used for reciprocal transplant and environmental manipulation studies.

Site	**Code**	**Latitude**	**Longitude**
University of Louisiana Ecology Center	LAF	30.304773	-92.012401
Center for Biological Field Studies	HUN	30.742815	-95.476881
TAMU Ecology and Natural Resource Teaching Area	COL	30.569302	-96.366324
UT Stengl Lost Pines Field Station	BAS	30.092343	-97.172442
1335 Medina Hwy, Kerrville, TX	KER	30.028437	-99.142888
Brackenridge Field Laboratory	BFL	30.284866	-97.780675
TAMU Sonora Station	SON	30.273467	-100.561463

Table S3: Comparison of candidate models (quadratic vs. linear) for each vital rate based on leave-one-out cross-validation (LOOCV). ELPD difference is relative to the best model.

Vital rate	Model type	ELPD difference	SE difference
Survival	Quadratic	0	0.000
Survival	Linear	-1.230	0.947
Growth	Quadratic	0	0.000
Growth	Linear	-0.544	0.924
Flowering	Quadratic	0	0.000
Flowering	Linear	-0.763	0.956
Spikelet	Quadratic	0	0.000
Spikelet	Linear	-0.097	0.957

Table S4: Posterior summaries of Δ ($E^{+} - E^{-}$) for survival at precipitation quantiles for each species and herbivore exclusion treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	P($\Delta > 0$)
Agrostis hyemalis	Unfenced	765.76	-0.01055	-0.12081	0.07964	0.3723
Agrostis hyemalis	Unfenced	848.28	-0.00814	-0.10368	0.07597	0.3980
Agrostis hyemalis	Unfenced	1113.25	0.00173	-0.07149	0.07904	0.5239
Agrostis hyemalis	Unfenced	1644.36	0.02230	-0.06632	0.14693	0.6981
Agrostis hyemalis	Unfenced	2163.28	0.03508	-0.08410	0.21801	0.7237
Agrostis hyemalis	Fenced	765.76	-0.04897	-0.17775	0.01302	0.0961
Agrostis hyemalis	Fenced	848.28	-0.04477	-0.15803	0.01516	0.1076
Agrostis hyemalis	Fenced	1113.25	-0.02642	-0.11372	0.03630	0.2266
Agrostis hyemalis	Fenced	1644.36	0.01141	-0.08847	0.12851	0.5950
Agrostis hyemalis	Fenced	2163.28	0.03789	-0.08484	0.21422	0.7222
Elymus virginicus	Unfenced	765.76	-0.02574	-0.19555	0.13388	0.3799
Elymus virginicus	Unfenced	848.28	-0.01565	-0.17000	0.13230	0.4250
Elymus virginicus	Unfenced	1113.25	0.01937	-0.10368	0.14589	0.6070
Elymus virginicus	Unfenced	1644.36	0.07749	-0.06409	0.22970	0.8217
Elymus virginicus	Unfenced	2163.28	0.10622	-0.07003	0.31076	0.8445
Elymus virginicus	Fenced	765.76	-0.12411	-0.28674	0.03254	0.0992
Elymus virginicus	Fenced	848.28	-0.09198	-0.23593	0.04709	0.1418
Elymus virginicus	Fenced	1113.25	-0.00179	-0.11213	0.11041	0.4882
Elymus virginicus	Fenced	1644.36	0.08855	-0.02165	0.24020	0.9069
Elymus virginicus	Fenced	2163.28	0.11327	-0.00261	0.32611	0.9456
Poa autumnalis	Unfenced	765.76	-0.00979	-0.12238	0.02097	0.1853
Poa autumnalis	Unfenced	848.28	-0.01778	-0.14683	0.02450	0.1735
Poa autumnalis	Unfenced	1113.25	-0.05636	-0.20042	0.03503	0.1429
Poa autumnalis	Unfenced	1644.36	-0.02891	-0.17290	0.07018	0.2804
Poa autumnalis	Unfenced	2163.28	-0.00174	-0.13201	0.07681	0.4319
Poa autumnalis	Fenced	765.76	0.00399	-0.03466	0.09583	0.6685
Poa autumnalis	Fenced	848.28	0.00739	-0.04282	0.11317	0.6803
Poa autumnalis	Fenced	1113.25	0.02557	-0.06906	0.15496	0.7063
Poa autumnalis	Fenced	1644.36	0.01678	-0.09915	0.15764	0.6297
Poa autumnalis	Fenced	2163.28	0.00181	-0.09727	0.12122	0.5663

Table S5: Posterior summaries of Δ ($E^+ - E^-$) for growth across precipitation quantiles for each species and herbivore treatment. Positive median values indicate beneficial effects of endophytes; negative medians indicate costly effects.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
Agrostis hyemalis	Unfenced	773.91	0.37006	-0.00924	0.75437	0.9456
Agrostis hyemalis	Unfenced	1075.09	0.37940	0.10078	0.65111	0.9866
Agrostis hyemalis	Unfenced	1316.25	0.38541	0.11668	0.64408	0.9915
Agrostis hyemalis	Unfenced	1979.35	0.39167	0.01393	0.77022	0.9539
Agrostis hyemalis	Unfenced	2163.28	0.39436	-0.02748	0.81310	0.9400
Agrostis hyemalis	Fenced	773.91	0.07955	-0.23563	0.39683	0.6612
Agrostis hyemalis	Fenced	1075.09	0.13909	-0.10277	0.37262	0.8279
Agrostis hyemalis	Fenced	1316.25	0.17455	-0.07319	0.41710	0.8781
Agrostis hyemalis	Fenced	1979.35	0.24676	-0.11528	0.60922	0.8733
Agrostis hyemalis	Fenced	2163.28	0.26153	-0.13527	0.66118	0.8657
Elymus virginicus	Unfenced	773.91	0.25869	-0.11938	0.62862	0.8693
Elymus virginicus	Unfenced	1075.09	0.24977	-0.02146	0.51978	0.9343
Elymus virginicus	Unfenced	1316.25	0.24421	0.00389	0.48413	0.9530
Elymus virginicus	Unfenced	1979.35	0.23379	-0.05653	0.52131	0.9096
Elymus virginicus	Unfenced	2163.28	0.23175	-0.08415	0.54666	0.8877
Elymus virginicus	Fenced	773.91	0.03164	-0.30420	0.37363	0.5644
Elymus virginicus	Fenced	1075.09	0.06458	-0.16044	0.29000	0.6825
Elymus virginicus	Fenced	1316.25	0.08512	-0.10777	0.27375	0.7682
Elymus virginicus	Fenced	1979.35	0.12472	-0.12987	0.37637	0.7903
Elymus virginicus	Fenced	2163.28	0.13352	-0.15046	0.41561	0.7788
Poa autumnalis	Unfenced	773.91	-0.27505	-0.74845	0.20422	0.1688
Poa autumnalis	Unfenced	1075.09	-0.14041	-0.44570	0.16567	0.2237
Poa autumnalis	Unfenced	1316.25	-0.05910	-0.29128	0.17786	0.3413
Poa autumnalis	Unfenced	1979.35	0.10761	-0.15864	0.37505	0.7388
Poa autumnalis	Unfenced	2163.28	0.14420	-0.16050	0.44856	0.7749
Poa autumnalis	Fenced	773.91	0.39049	-0.05133	0.83526	0.9270
Poa autumnalis	Fenced	1075.09	0.16142	-0.12313	0.45470	0.8212
Poa autumnalis	Fenced	1316.25	0.02003	-0.20891	0.25959	0.5587
Poa autumnalis	Fenced	1979.35	-0.26222	-0.54759	0.03589	0.0723
Poa autumnalis	Fenced	2163.28	-0.32245	-0.64896	0.01200	0.0558

Table S6: Posterior summaries of Δ ($E^{+} - E^{-}$) for flowering output at precipitation quantiles for each species and herbivore exclusion treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	P($\Delta > 0$)
Agrostis hyemalis	Unfenced	773.91	-0.6824	-0.6904	0.7822	0.4889
Agrostis hyemalis	Unfenced	1075.09	0.0286	-0.6483	0.9444	0.6018
Agrostis hyemalis	Unfenced	1316.25	0.4663	-0.6218	1.2633	0.6909
Agrostis hyemalis	Unfenced	1979.35	1.3488	-1.2041	4.0378	0.7552
Agrostis hyemalis	Unfenced	2163.28	1.5409	-1.4924	5.3973	0.7522
Agrostis hyemalis	Fenced	773.91	-0.6824	-0.7351	0.9077	0.5238
Agrostis hyemalis	Fenced	1075.09	0.0286	-0.6302	1.2286	0.6696
Agrostis hyemalis	Fenced	1316.25	0.4663	-0.6108	1.7763	0.7609
Agrostis hyemalis	Fenced	1979.35	1.3488	-1.3208	5.6835	0.7965
Agrostis hyemalis	Fenced	2163.28	1.5409	-1.7235	7.3637	0.7922
Elymus virginicus	Unfenced	773.91	-0.6824	-0.4386	0.2723	0.3289
Elymus virginicus	Unfenced	1075.09	0.0286	-0.3390	0.3120	0.4611
Elymus virginicus	Unfenced	1316.25	0.4663	-0.2792	0.3903	0.5915
Elymus virginicus	Unfenced	1979.35	1.3488	-0.2491	0.8798	0.7902
Elymus virginicus	Unfenced	2163.28	1.5409	-0.2647	1.0691	0.8082
Elymus virginicus	Fenced	773.91	-0.6824	-0.8003	0.7325	0.4283
Elymus virginicus	Fenced	1075.09	0.0286	-0.6057	0.5997	0.4688
Elymus virginicus	Fenced	1316.25	0.4663	-0.4972	0.5628	0.5208
Elymus virginicus	Fenced	1979.35	1.3488	-0.6334	1.0660	0.6272
Elymus virginicus	Fenced	2163.28	1.5409	-0.7473	1.3341	0.6318
Poa autumnalis	Unfenced	773.91	-0.6824	-0.1587	0.7919	0.7062
Poa autumnalis	Unfenced	1075.09	0.0286	-0.0711	0.9392	0.8638
Poa autumnalis	Unfenced	1316.25	0.4663	0.0137	1.3447	0.9629
Poa autumnalis	Unfenced	1979.35	1.3488	0.0793	6.8120	0.9926
Poa autumnalis	Unfenced	2163.28	1.5409	0.0645	10.6951	0.9858
Poa autumnalis	Fenced	773.91	-0.6824	0.0907	5.6618	0.9981
Poa autumnalis	Fenced	1075.09	0.0286	0.2492	4.3096	0.9998
Poa autumnalis	Fenced	1316.25	0.4663	0.3164	4.7790	0.9999
Poa autumnalis	Fenced	1979.35	1.3488	0.1612	11.8419	0.9976
Poa autumnalis	Fenced	2163.28	1.5409	0.0888	15.3801	0.9848

Table S7: Posterior summaries of Δ ($E^+ - E^-$) at precipitation quantiles for each species and herbivore exclusion treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
Elymus virginicus	Unfenced	1113.25	0.1040	-4.4921	2.4788	0.3213
Elymus virginicus	Unfenced	1276.99	0.4008	-3.6975	2.7993	0.4012
Elymus virginicus	Unfenced	1644.36	0.9477	-2.5664	3.8884	0.6089
Elymus virginicus	Unfenced	2163.28	1.5409	-2.0652	6.1715	0.7733
Elymus virginicus	Unfenced	2182.71	1.5603	-2.0610	6.2851	0.7760
Elymus virginicus	Fenced	1113.25	0.1040	-4.2244	1.7020	0.2288
Elymus virginicus	Fenced	1276.99	0.4008	-3.0122	2.0977	0.3714
Elymus virginicus	Fenced	1644.36	0.9477	-1.2591	3.8036	0.7853
Elymus virginicus	Fenced	2163.28	1.5409	-0.4058	7.9946	0.9282
Elymus virginicus	Fenced	2182.71	1.5603	-0.3886	8.1667	0.9288
Poa autumnalis	Unfenced	1113.25	0.1040	-3.2874	1.7757	0.3134
Poa autumnalis	Unfenced	1276.99	0.4008	-2.7023	1.8008	0.3654
Poa autumnalis	Unfenced	1644.36	0.9477	-1.7415	2.3862	0.5832
Poa autumnalis	Unfenced	2163.28	1.5409	-1.9577	5.4441	0.7572
Poa autumnalis	Unfenced	2182.71	1.5603	-1.9811	5.5917	0.7592
Poa autumnalis	Fenced	1113.25	0.1040	0.3203	6.9989	0.9668
Poa autumnalis	Fenced	1276.99	0.4008	0.6863	6.6041	0.9828
Poa autumnalis	Fenced	1644.36	0.9477	0.8105	6.7216	0.9873
Poa autumnalis	Fenced	2163.28	1.5409	-0.9449	8.6686	0.8957
Poa autumnalis	Fenced	2182.71	1.5603	-1.0369	8.7690	0.8906

S.1.3 Supporting Figures

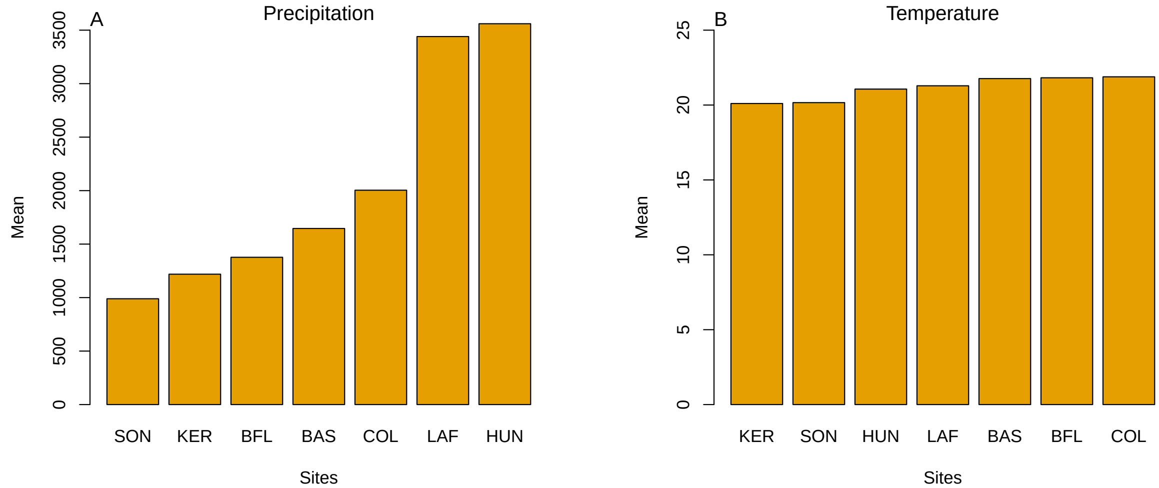

Fig S1: Abiotic conditions across the seven study sites. (A) Mean annual precipitation showing the pronounced east–west aridity gradient from mesic Louisiana to arid Texas. (B) Mean annual temperature across sites, which does not differ much. Panels highlight that precipitation, rather than temperature, constitutes the primary abiotic gradient in this system.

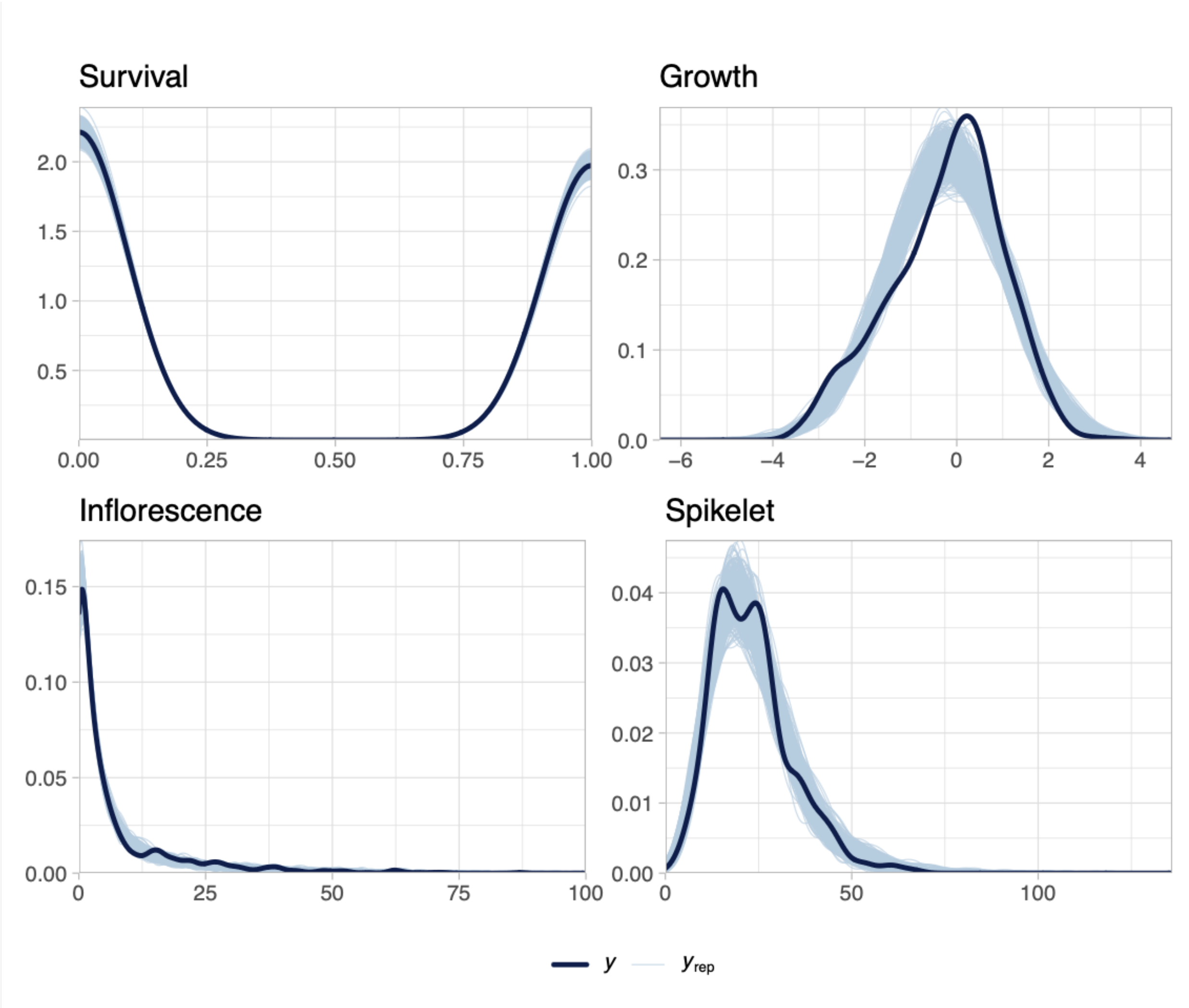

Fig S2: Posterior predictive checks for each vital rate. Dark blue lines show observed data (y), and light blue lines show replicated datasets generated from posterior predictions (y_{rep}). The strong overlap across survival, growth, flowering, and spikelet production indicates adequate model fit for all vital rates.

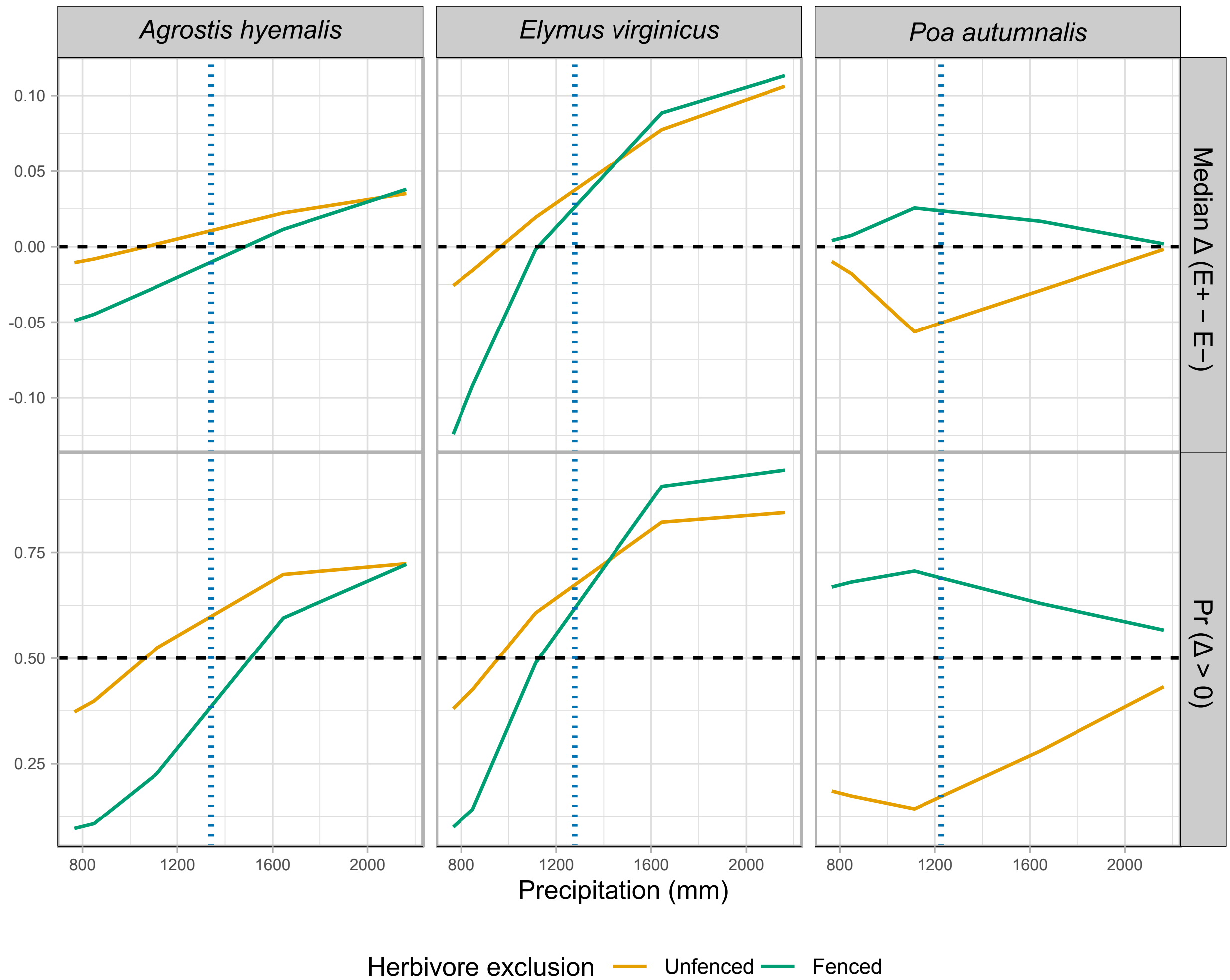

Fig S3: Predicted effects of endophyte presence (E^+) versus absence (E^-) on survival across precipitation gradients for three cool-season grasses. Solid lines show the median Δ ($E^+ - E^-$),or the posterior probability that $\Delta > 0$ under fenced and unfenced conditions. Horizontal dashed lines indicate no effect ($\Delta = 0$) or a 50% probability of benefit. Vertical dashed lines represent the precipitation values at the most westerly natural occurrence of each species, based on GBIF records.

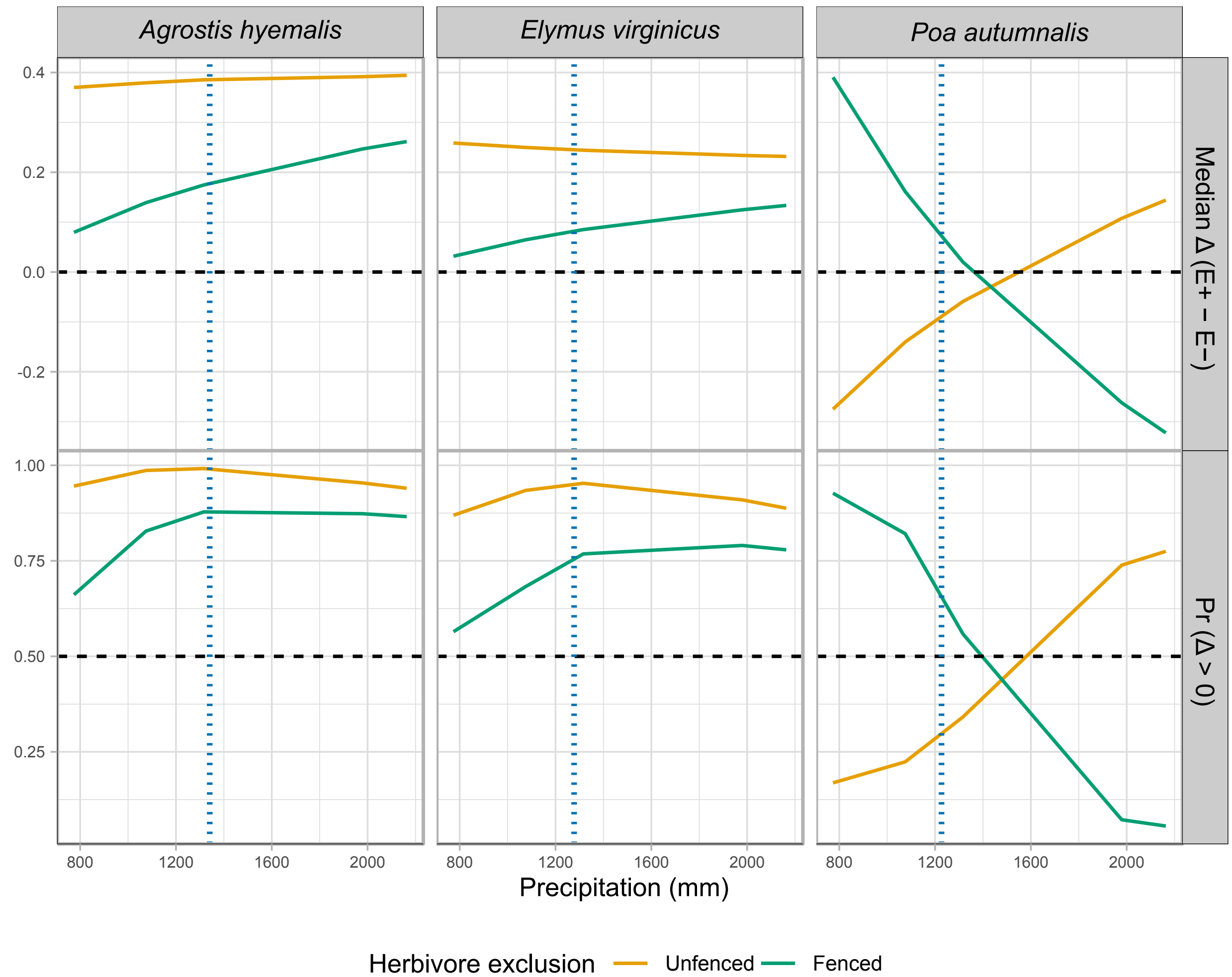

Fig S4: Effects of precipitation and endophyte presence on growth across three cool-season grass species. Lines show the predicted difference in growth between endophyte-present (E^{+}) and endophyte-absent (E^{-}) plants (Median Δ, top row) and the posterior probability that this difference is positive ($Pr(\Delta > 0)$, bottom row) under fenced and unfenced treatments. Predictions were calculated at the 10th, 25th, 50th, 75th, and 90th percentiles of precipitation observed in the study. Horizontal dashed lines indicate $\Delta = 0$ (top row) and $Pr(\Delta > 0) = 0.5$ (bottom row). Vertical dashed lines represent the precipitation values at the most westerly natural occurrence of each species, based on GBIF records.

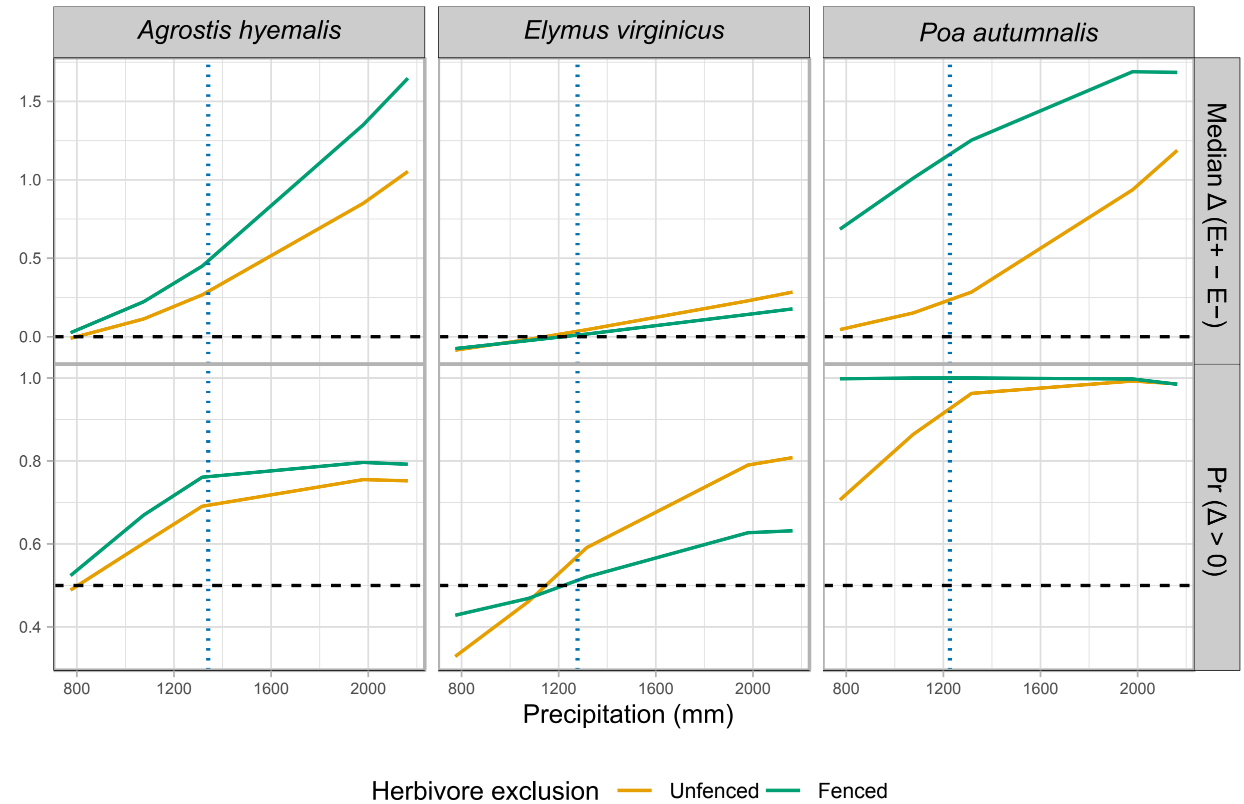

Fig S5: Effects of endophyte infection on flowering output across a precipitation gradient for three cool-season grass species. Lines show the median posterior difference Δ $(E^+ - E^-)$ and the posterior probability that $\Delta > 0$ under fenced (herbivores excluded) and unfenced (herbivores present) conditions. Horizontal dashed lines indicate $\Delta = 0$ for median differences and $\Pr(\Delta > 0) = 0.5$. Endophyte effects were generally positive for *Agrostis hyemalis* and *Poa autumnalis*, but for *Elymus virginicus* benefits were context-dependent, with costs at lower precipitation

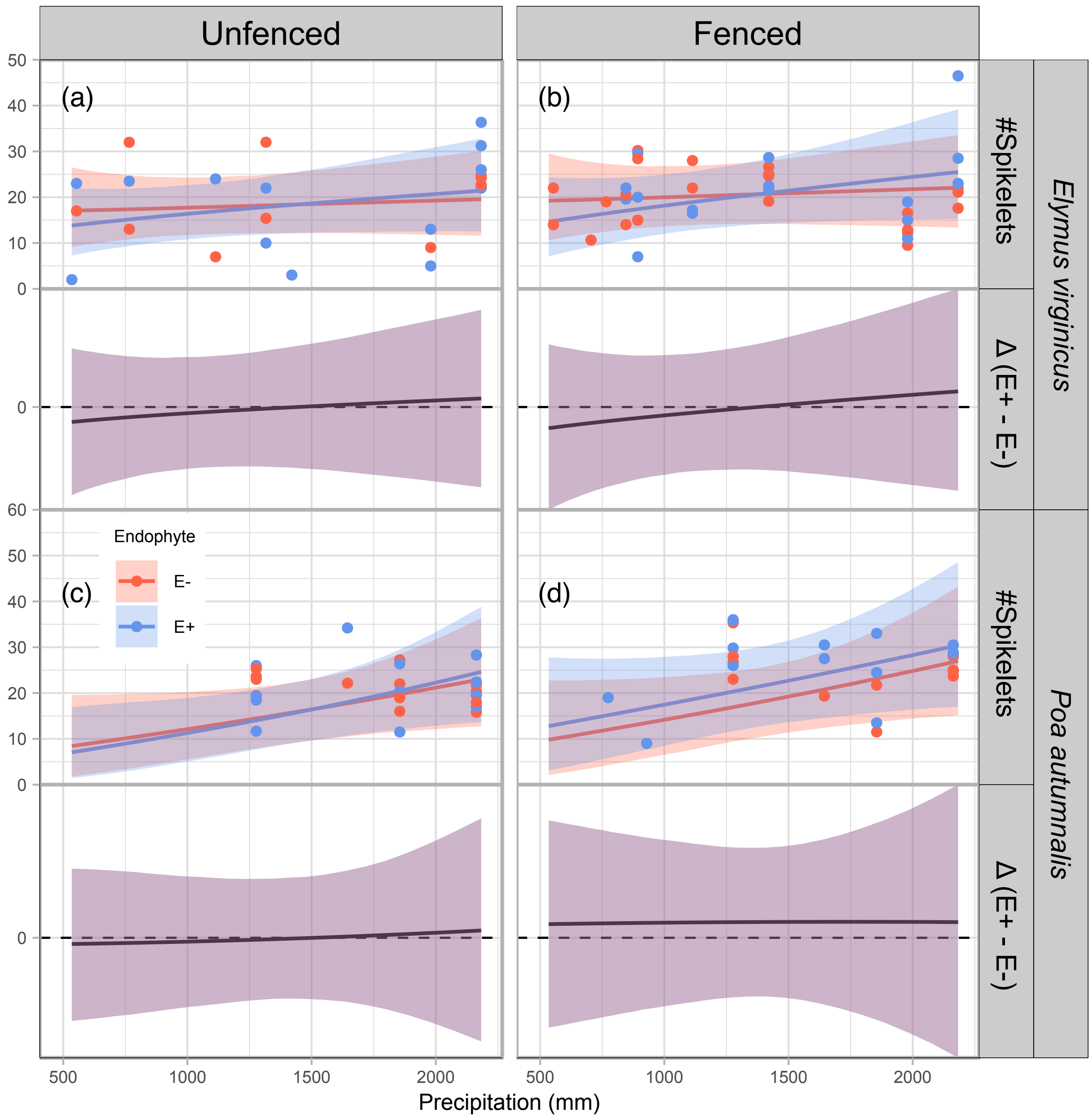

Fig S6: Predicted number of spikelets (upper panels) and endophyte effect ($\Delta = E^{+} - E^{-}$, bottom panels) across precipitation gradients. Lines show posterior means for E^{+} (blue) and E^{-} (red), with shading indicating 90% credible intervals. Points represent mean observed spikelets per plot, slightly jittered for visualization. Spikelets were measured for only two species (see Methods for details). The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

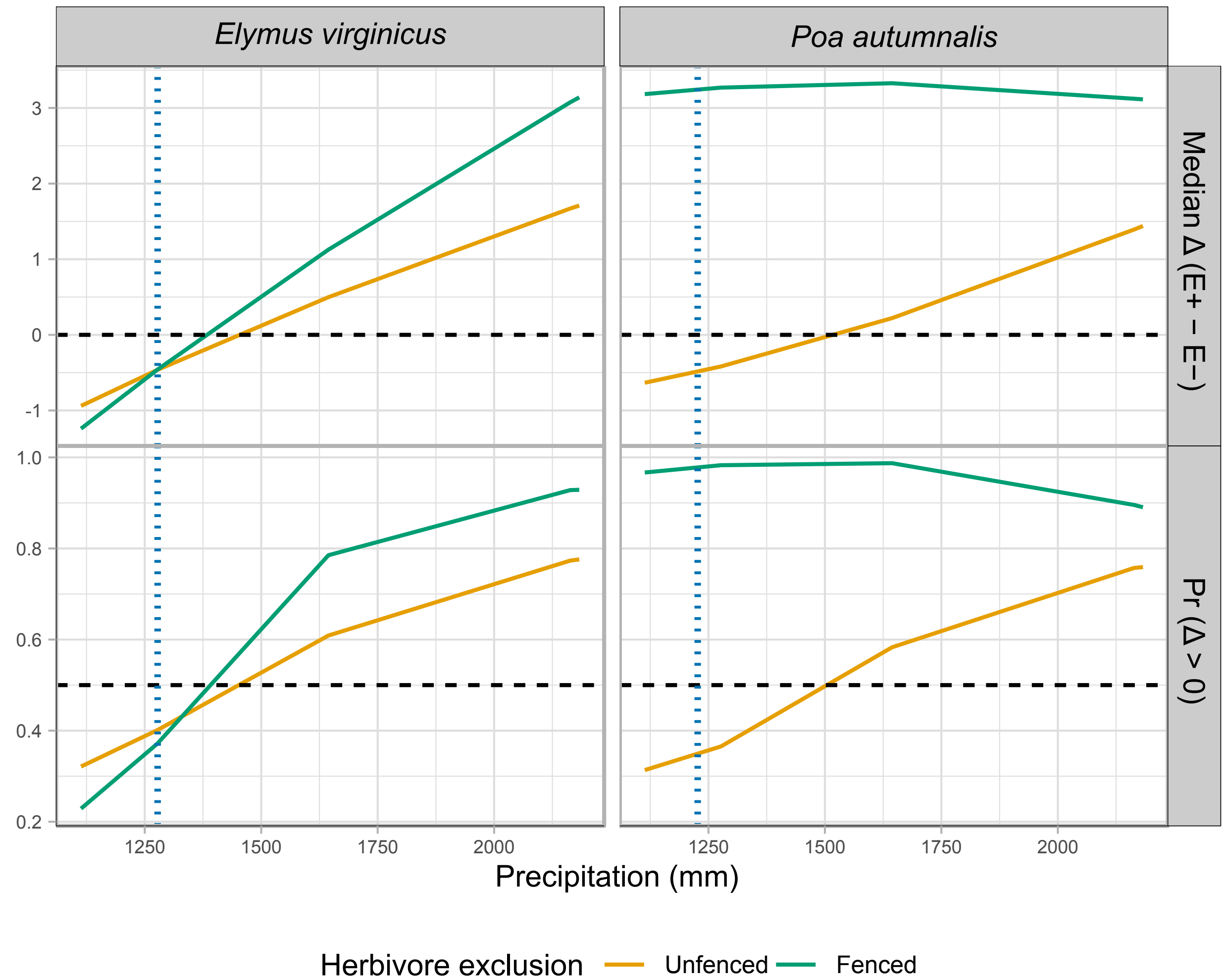

Fig S7: Effects of precipitation and endophyte presence on spikelet (reproductive output) for three cool-season grass species. Median Δ ($E^+ - E^-$) and posterior probability $Pr(\Delta > 0)$ are shown across precipitation quantiles. Herbivore exclusion treatments are indicated by color (Fenced = herbivores excluded, Unfenced = herbivores had access). Horizontal dashed lines indicate $\Delta = 0$ for medians and $Pr(\Delta > 0) = 0.5$ for probabilities. Facets separate metrics and species.